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Multi-Currency Portfolio Optimization with General Risk Measures

Master Thesis

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Acknowledgments

Abstract

Multi-currency portfolio construction requires managing asset and currency exposures together, yet it remains unclear whether these decisions are best made simultaneously or sequentially as the investable universe grows. This thesis builds on the regularized multi-currency Expected Shortfall framework of Lucescu and Ulrych (2026), in which portfolios are formed by minimizing a risk objective subject to L^1 and L^2 penalties, with currency exposures either optimized jointly with assets or handled through a separate overlay. First we replicate the framework's core strategies, then extend it along two dimensions, the size of the investable universe, and the choice of objective function, implementing both Expected Shortfall and mean-variance formulations of the joint and overlay optimizers.

Across both objectives, the joint optimizer's out-of-sample performance deteriorates relative to the separate overlay as the universe expands. This motivates the central question of the thesis, is this deterioration a structural property of joint optimization, or is it modulated by the choice of objective function? To answer it, we develop a general regularized optimization framework that admits arbitrary objective functions within a single, common formulation. We validate the framework by showing that it exactly reproduces the existing Expected Shortfall and mean-variance optimizers, and then use it to measure how the rate of the joint optimizer's breakdown varies as dimensionality increases.

Sentence on results.

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Chapter 1

Introduction

1.1 Motivation: International Diversification, Currency Risk, and Tail Risk

1.2 Contributions

1.3 Thesis Structure

Chapter 2

Literature Review

Cite the following: International allocation and currency risk management, Currency overlay vs. joint optimization, Downside risk measures and Expected Shortfall, Instability of ES optimization and the role of regularization, Generalized objectives and decision-focused / end-to-end learning.

In this chapter, we review the relevant literature on multi-currency portfolio optimization. We discuss the contributions and methodologies in the field, highlighting how our work builds upon and extends existing research.

The core of our analysis builds on two recent studies that apply regularization techniques to international asset allocation. Burkhardt and Ulrych (2023) introduce a joint optimization approach within a mean-variance framework. By using L1 and L2 regularization, they show that jointly optimizing assets and currencies leads to better out-of-sample performance than the standard industry practice of using a separate currency overlay.

Extending this framework, Lucescu and Ulrych (2026) consider an Expected Shortfall optimization framework to better manage downside tail risk. To fix the numerical instability common in Expected Shortfall minimization, they apply similar regularization methods. Their findings reveal a dimensional trade-off. Joint optimization works best in small asset universes, but the separate currency overlay approach becomes more effective for larger universes due to its built-in dimensionality reduction.

Reproducing and comparing these two baseline frameworks serves as the starting point for the extensions proposed in this thesis.

Chapter 3

Optimization Framework

This chapter develops the optimization framework used throughout the thesis, building up from a model of multi-currency portfolio returns to the risk objectives, the regularization that stabilizes them, and the two formulations through which asset and currency decisions are combined. The derivations follow the methodology presented in [25] and [24], use simple returns, and assume that rebalancing occurs at the end of each investment period. We begin in Section 3.1 with a model-free representation of one-period portfolio returns for an investor who holds assets denominated in multiple currencies and can take currency forward positions for hedging, decomposing the unhedged return of a foreign asset into its local, currency, and cross-currency components, introducing the currency-mapping matrix that links assets to their implicit currency exposures, and expressing the hedged portfolio return in terms of net currency exposures.

With the return model in place, we formulate the optimization problem. Section 3.3 states the classical mean-variance criterion, and Section 3.4 introduces Expected Shortfall as a coherent downside-risk measure, together with the Rockafellar–Uryasev representation that casts its minimization as a tractable linear program. Both are presented as instances of a common mean–risk objective, so that the framework treats the risk measure as a swappable component rather than a fixed choice. Section **??** then introduces the L_1 and L_2 penalties and the leverage constraint used to stabilize the optimization and improve out-of-sample performance.

Finally, Sections **??** and **??** assemble these elements into the two approaches to multi-currency allocation studied throughout the thesis: the separate (currency overlay) formulation, in which asset and currency decisions are made sequentially, and the joint formulation, in which they are optimized simultaneously. The detailed derivations of the resulting quadratic programs are found in Appendix A and Appendix B.

3.1 Model-Free Multi-Currency Portfolio Returns

3.1.1 Unhedged Portfolio Returns

We consider an international investor who allocates their wealth across N assets denominated in up to $M + 1$ currencies. We fix a discrete time index $t \in \{0, 1, 2, \dots\}$ and one investment period from t to $t+1$. Let the investor's *domestic currency* be indexed by 0 and let the remaining M foreign currencies be indexed by $\mathcal{C} \coloneqq \{1, 2, \dots, M\}$.

Let $P_{i,t} > 0$ be the price of asset i at time t in its local currency $c_i \in \mathcal{C} \cup \{0\}$. Define the

corresponding *simple local-currency asset return* over $(t, t{+}1]$ as

$$R_{i,t+1} \coloneqq \frac{P_{i,t+1} - P_{i,t}}{P_{i,t}}.$$

Thus $P_{i,t+1} = P_{i,t}(1 + R_{i,t+1})$.

For each foreign currency $c \in \mathcal{C}$, let $S_{c,t} > 0$ denote the exchange rate at time t, quoted as units of domestic currency per one unit of foreign currency c. The associated *simple currency return* over $(t, t{+}1]$ is

$$e_{c,t+1} \coloneqq \frac{S_{c,t+1} - S_{c,t}}{S_{c,t}}.$$

Equivalently, $S_{c,t+1} = S_{c,t}(1+e_{c,t+1})$. For the domestic currency, we set $S_{0,t} \equiv 1$ and $e_{0,t+1} \equiv 0$.

The domestic-currency value of holding one unit of asset i at time t is $P_{i,t}\, S_{c_i,t}$. Therefore, the *unhedged domestic-currency simple return* of asset i is

$$\begin{aligned} R^u_{i,t+1} &= \frac{P_{i,t+1}S_{c_i,t+1} - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= \frac{P_{i,t}(1 + R_{i,t+1})\, S_{c_i,t}(1 + e_{c_i,t+1}) - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= R_{i,t+1} + e_{c_i,t+1} + R_{i,t+1}\, e_{c_i,t+1}. \end{aligned} \tag{3.1}$$

Let $\boldsymbol{x}_t \in \mathbb{R}^N$ be the vector of asset portfolio weights chosen at time t and held until $t{+}1$, where $x_{i,t}$ is the fraction of total portfolio value invested in asset i at time t. We impose the standard budget constraint $\mathbf{1}^\mathsf{T}\boldsymbol{x}_t = 1$.

Assuming no rebalancing between times t and $t{+}1$, the domestic-currency portfolio's unhedged return is given by

$$R^u_{\mathcal{P},t+1} = \sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}.$$

3.1.2 Hedged Portfolio Returns

To manage currency risk, the investor can take positions in one-period currency forward contracts. For each foreign currency $c \in \mathcal{C}$, let $F_{c,t}$ be the forward rate quoted at time t for settlement at $t{+}1$, in units of domestic currency per unit of foreign currency c. We define the *forward premium* by

$$f_{c,t} \coloneqq \frac{F_{c,t} - S_{c,t}}{S_{c,t}}.$$

The investor hedges by selling foreign currency forward. Let $\phi_{c,t}$ denote the relative notional of the *short* forward position in currency c at time t, expressed as a fraction of total portfolio value, with the convention that $\phi_{c,t} > 0$ corresponds to hedging. Per unit of notional, the short position pays $f_{c,t} - e_{c,t+1}$ in domestic currency over the period. Adding these forward payoffs to the unhedged asset returns, the *hedged portfolio return* is

$$R^h_{\mathcal{P},t+1} = \underbrace{\sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}}_{R^u_{\mathcal{P},t+1}} + \sum_{c=1}^{M} \phi_{c,t}\,(f_{c,t} - e_{c,t+1}). \tag{3.2}$$

3.1.3 Matrix Representation and Decomposition

To express the returns in a compact form suitable for optimization, we define the currency incidence matrix $\boldsymbol{C} \in \{0,1\}^{N\times M}$ by

$$C_{i,c} \coloneqq \begin{cases} 1, & \text{if asset } i \text{ is denominated in foreign currency } c, \\ 0, & \text{otherwise.} \end{cases}$$

Each row of $\boldsymbol{C}$ has at most one nonzero entry, and the rows corresponding to domestic assets are identically zero. This allows us to define the portfolio's *implicit foreign-currency exposure* as

$$\boldsymbol{w}_t \coloneqq \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t \in \mathbb{R}^M, \qquad w_{c,t} = \sum_{i:\, c_i=c} x_{i,t},$$

such that $w_{c,t}$ is the total portfolio weight invested in assets denominated in currency c. The implicit domestic exposure follows residually as $w_{0,t} = 1 - \mathbf{1}_M^\mathsf{T}\boldsymbol{w}_t$, since the asset weights sum to one.

We express returns in vector form $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$, $\boldsymbol{e}_{t+1} \in \mathbb{R}^M$, and the forward premiums as $\boldsymbol{f}_t \in \mathbb{R}^M$. The product $\boldsymbol{C}\boldsymbol{e}_{t+1} \in \mathbb{R}^N$ aligns currency returns to assets, with ith component $e_{c_i,t+1}$ for foreign assets and 0 for domestic ones. Using (3.1), the vector of *unhedged domestic returns* across assets can be written as

$$\boldsymbol{R}^u_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}),$$

where $\odot$ denotes the Hadamard product.

Expressing the forward positions in vector form as $\boldsymbol{\phi}_t \in \mathbb{R}^M$, the hedged portfolio return (3.2) can be written as

$$R^h_{\mathcal{P},t+1} = \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + \boldsymbol{\phi}_t^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}). \tag{3.3}$$

Setting $\boldsymbol{\phi}_t = \boldsymbol{w}_t = \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t$ neutralizes the implicit currency exposure. Substituting this together with the unhedged return vector into (3.3), the *fully hedged portfolio return* is

$$\begin{aligned} R^{fh}_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + (\boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t)^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1})\Big] + \boldsymbol{x}_t^\mathsf{T}\boldsymbol{C}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}) + \boldsymbol{C}\boldsymbol{f}_t - \boldsymbol{C}\boldsymbol{e}_{t+1}\Big] \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1})\Big] \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^{fh}_{t+1}. \end{aligned}$$

The first-order currency terms $\pm\boldsymbol{x}_t^\mathsf{T}\boldsymbol{C}\boldsymbol{e}_{t+1}$ cancel, leaving the *fully hedged asset return vector*

$$\boldsymbol{R}^{fh}_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}).$$

More generally, define the *net currency exposure* $\boldsymbol{\psi}_t \coloneqq \boldsymbol{w}_t - \boldsymbol{\phi}_t$. Imposing $\sum_{c=0}^M \phi_{c,t} = 1$ gives $\sum_{c=0}^M \psi_{c,t} = 0$, so the currency overlay is a zero-investment portfolio. Substituting $\boldsymbol{\phi}_t = \boldsymbol{w}_t - \boldsymbol{\psi}_t$ into (3.3) expresses the hedged return in terms of net exposures,

$$\begin{aligned} R^h_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + (\boldsymbol{w}_t - \boldsymbol{\psi}_t)^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + \boldsymbol{w}_t^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) + \boldsymbol{\psi}_t^\mathsf{T}(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^{fh}_{t+1} + \boldsymbol{\psi}_t^\mathsf{T}(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t), \end{aligned} \tag{3.4}$$

which we adopt throughout the following portfolio optimization formulations, as it makes the asset weights $\boldsymbol{x}_t$ and the net currency exposures $\boldsymbol{\psi}_t$ the explicit variables of the optimization.

3.2 Mean–Risk Portfolio Optimization

3.3 Mean-Variance Optimization Formulation

3.4 Expected Shortfall Optimization Formulation

Chapter 4

Data and Implementation

In this chapter, we describe the dataset used for the empirical evaluation of the portfolio optimization strategies and the implementation of the backtesting framework. In the data section we detail the construction of the investable universe. We follow [24] to analyze asset and excess currency returns and volatillities. We further expand the analysis to different investable universe sizes, and look at the correlation dependence of asset and currency returns. In the implementation section we describe the modular structure of the backtesting framework, the workflow of the backtest, numerical considerations in the implementation, and the metrics used to evaluate potrfolio performance.

4.1 Data

The empirical evaluation of all portfolio optimization strategies in this thesis relies on a dataset of international equities, foreign exchange spot rates, and forward exchange rates sourced from Refinitiv Datastream. The the one-month U.S. Treasury bill rate used in the return and performance metric calculations is obtained from Kenneth R. French's online data library.[1] The datasets span from June 1990 to June 2023, covering several different market regimes, including the dot-com crash, the 2008 Global Financial Crisis, the European sovereign debt crisis, and the COVID-19 pandemic.

We construct the investable universe from 14 global equity markets: Australia, Canada, Denmark, Norway, Sweden, Hong Kong, Japan, New Zealand, Singapore, South Africa, Switzerland, the Eurozone, the United Kingdom, and the United States. The universe is rebuilt at the beginning of each year from the largest stocks by market capitalization at the time (a rolling-window approach that mitigates survivorship bias). At the start of each year we rank stocks within each market and restrict the following year's investments to that set. To study how the universe size affects the optimization approaches considered, we form three cross-sections, comprising the top 10, top 50, and top 100 stocks per market. Because the constituents turn over each year, each universe spans more distinct stocks over the full sample than are held on any single date, 407 for the top 10, 1705 for the top 50, and 3189 for the top 100, respectively.

The US dollar serves as the base currency throughout our analysis. For each stock we compute fully hedged returns, and for each currency the excess currency return, following the model-free return representation in Chapter 3.

[1] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Economies	Assets (%)						Currencies (%)	
	Top 10		Top 50		Top 100			
	Return	Vol.	Return	Vol.	Return	Vol.	Excess Return	Vol.
Australia	4.57	29.94	3.10	37.72	-1.88	58.00	1.31	11.59
Canada	9.10	32.89	6.58	40.24	1.86	59.24	-0.03	7.63
Denmark	8.35	34.05	3.04	39.00	-3.47	48.52	-0.08	9.68
Eurozone	5.09	32.24	5.05	34.01	3.25	35.46	-0.40	9.58
Hong Kong	4.42	37.59	-0.15	45.77	-3.48	52.59	-0.16	0.56
Japan	5.37	33.69	4.41	34.01	4.09	35.38	-2.07	10.47
New Zealand	2.88	29.93	-1.13	37.16	-4.56	41.49	2.48	11.36
Norway	3.17	43.30	-1.59	50.29	-2.61	53.26	-0.36	11.73
Singapore	3.32	35.43	-1.99	54.57	-4.84	62.12	0.18	5.35
South Africa	3.53	37.72	1.01	43.84	-4.39	53.94	1.22	15.18
Sweden	9.54	34.50	6.92	42.50	5.37	47.65	-1.16	11.50
Switzerland	6.38	29.04	5.67	30.32	3.97	33.65	-0.12	10.99
United Kingdom	2.76	31.50	2.63	34.71	1.29	36.22	0.12	9.42
United States	12.59	33.58	7.99	39.07	6.86	42.84	0.00	0.00

Table 4.1: Summary statistics of fully hedged equity returns and excess currency returns by economy. Asset statistics are reported for three investable universes (the top 10, 50, and 100 stocks per economy). Excess currency returns are defined as the currency return net of the forward premium. All numbers are reported as annualized percentages, from the perspective of a USD-based investor.

Table 4.1 reports annualized average returns and volatilities for the fully hedged equity returns and excess currency returns, by economy. For each economy, the asset figures are the cross-sectional average of the daily returns of its individual stocks.

We can see that as the investable universe becomes larger, the average returns tend to fall. In the top 10 universe every economy has a positive average return, ranging from 2.76% (the United Kingdom) to 12.59% (the United States). In the top 50 universe, the range drops downward, from −1.99% (Singapore) to 7.99% (the United States), and in the top 100 universe the range goes from −4.84% (Singapore) to 6.86% (the United States), with many economies now having negative average returns. The trend reflects the firm size, the largest stocks post the strongest returns, and the smaller firms added pull the average down. Volatilities exhibit the opposite pattern, rising with universe size, as smaller, more volatile stocks join the investable universe. Switzerland is the least volatile market in every case, its volatility rises from 29.04% to 33.65%, and the upper end rises from 43.30% (Norway) in the top 10 to 62.12% (Singapore) in the top 100.

Excess currency returns range between −2.07% for the Japanese yen and 2.48% for the New Zealand dollar, and excess currency volatilities range from 0.56% for the Hong Kong dollar, whose peg to the US dollar keeps it stable, up to 15.18% for the South African rand.

To quantify the dependence between asset and currency returns, we compute Pearson correlations between economy-level fully hedged equity returns and excess currency returns. For each economy we form an equally weighted daily portfolio of the fully hedged returns of its constituent stocks, resetting the constituents at each annual universe reconstitution. Correlating each economy's series with each currency's excess return over the full sample gives a 14×13 correlation matrix. The US dollar is dropped from the currency axis, since its excess return is zero by construction. We repeat this for the three universe sizes. The resulting heatmaps are shown in Figure 4.1, Figure 4.2, and Figure 4.3.

Across all three universes the correlations range from −0.2 to 0.4, and remain mostly positive. The pattern is intuitive and confirms well known empirical findings. The strongest positive

correlations belong to commodity and risk-sensitive currencies such as the AUD, CAD, NZD and ZAR, whose equity markets and currencies rise together in good times and fall together in bad [4, 14, 16]. The JPY and CHF are the only negatively correlated currencies with most other economies (ranging from -0.2 to 0.03), in line with their role as safe havens that strengthen when equities fall [21]. The EUR, DKK, and HKD are minimally correlated, reflecting the HKD's peg to the US dollar, and the DKK's peg to the EUR. This split between risky and safe-haven currencies is well documented [3, 21].

This is the dependence structure the overlay optimization approach discards. It sets asset weights first, using only equity information, and adds the currency hedge afterwards, whereas the joint model can act on it directly. The structure barely changes across universe sizes, and this matters for the results that follow. A shift in the relative performance of the joint and overlay approaches in Chapter 5 should therefore be attributed to estimation error rather than a change in the underlying dependence structure.

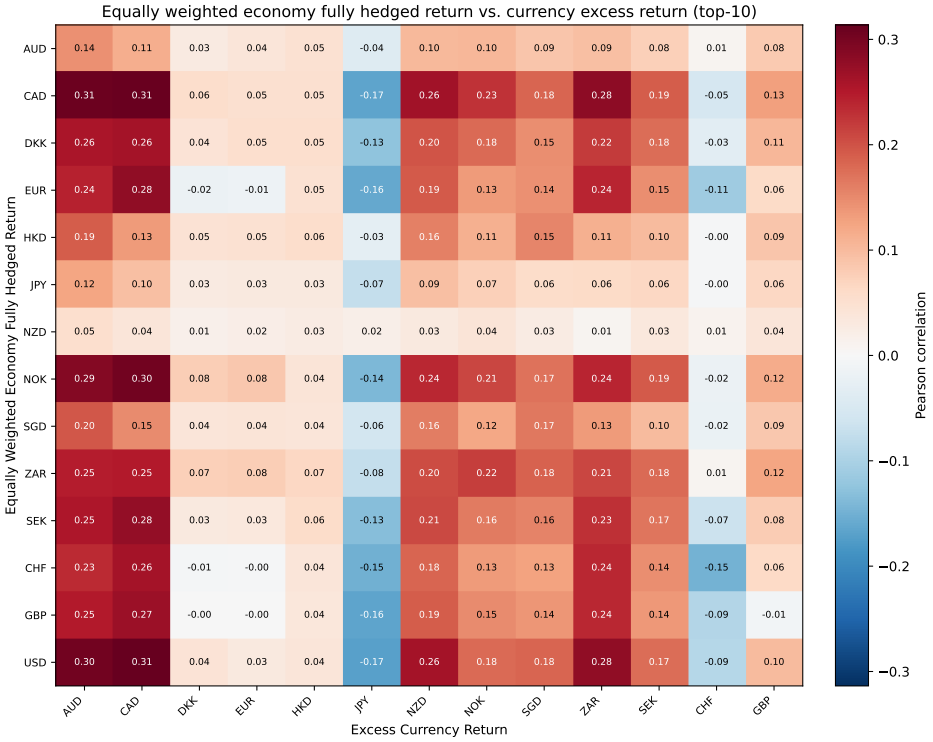


Figure 4.1: Pearson correlation between equally weighted economy fully hedged returns and excess currency returns, across the top 10 economies. The US dollar is excluded from the currency axis, since it serves as the base currency and therefore its excess return is zero.

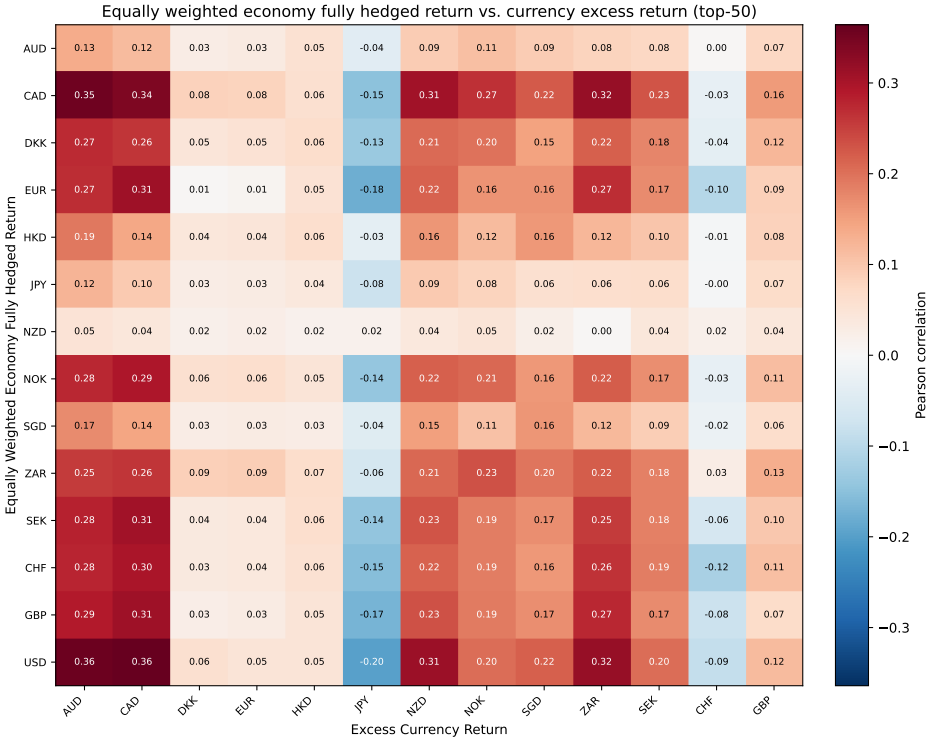


Figure 4.2: Pearson correlation between equally weighted economy fully hedged returns and excess currency returns, across the top 50 economies. The US dollar is excluded from the currency axis, since it serves as the base currency and therefore its excess return is zero.

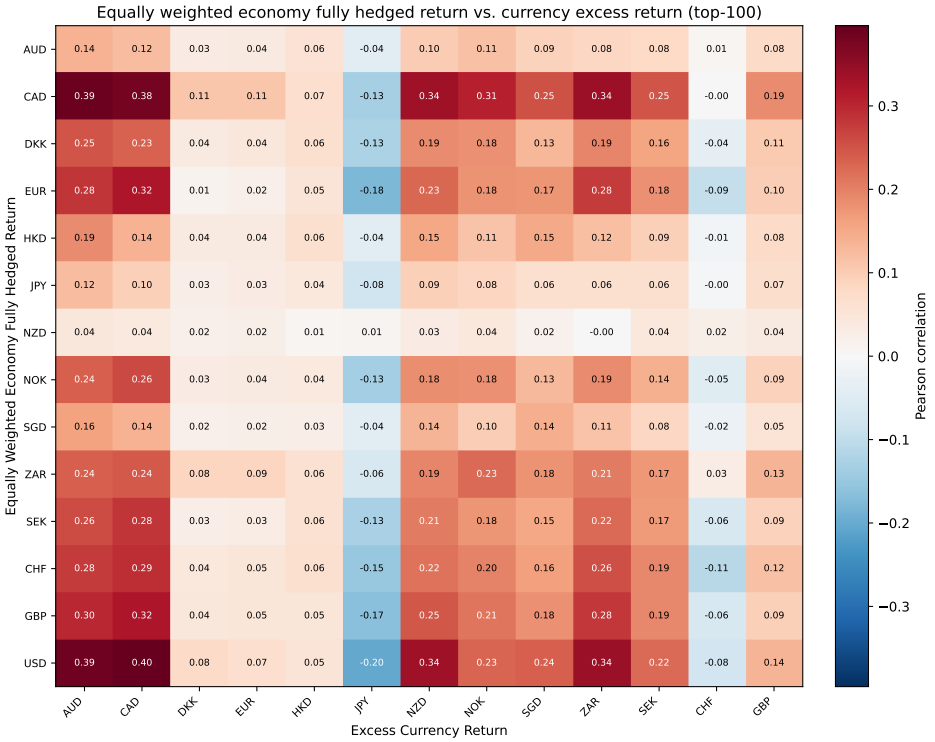


Figure 4.3: Pearson correlation between equally weighted economy fully hedged returns and excess currency returns, across the top 100 economies. The US dollar is excluded from the currency axis, since it serves as the base currency and therefore its excess return is zero.

4.2 Backtesting Framework

In Chapter 5 we evaluate a range of optimization strategies under different objectives (Mean-Variance, Expected Shortfall, Mean Absolute Deviation), their treatment of currency exposure (overlay or joint), and the size of the investable universe. To keep these comparisons fair, the backtesting framework separates between the orchestration of the backtest and the optimization logic it evaluates.[2] The main backtesting loop that ingests data, rolls the estimation window forward, selects hyperparameters, and records performance is fixed, while the optimizer that produces portfolio weights is a swappable component. Any optimizer that conforms to a common interface is therefore evaluated under the same conditions. The framework is composed of three main components, a data handler, the backtester, and a family of optimizers under a common abstract base class, whose interaction is summarized in Figure 4.4.

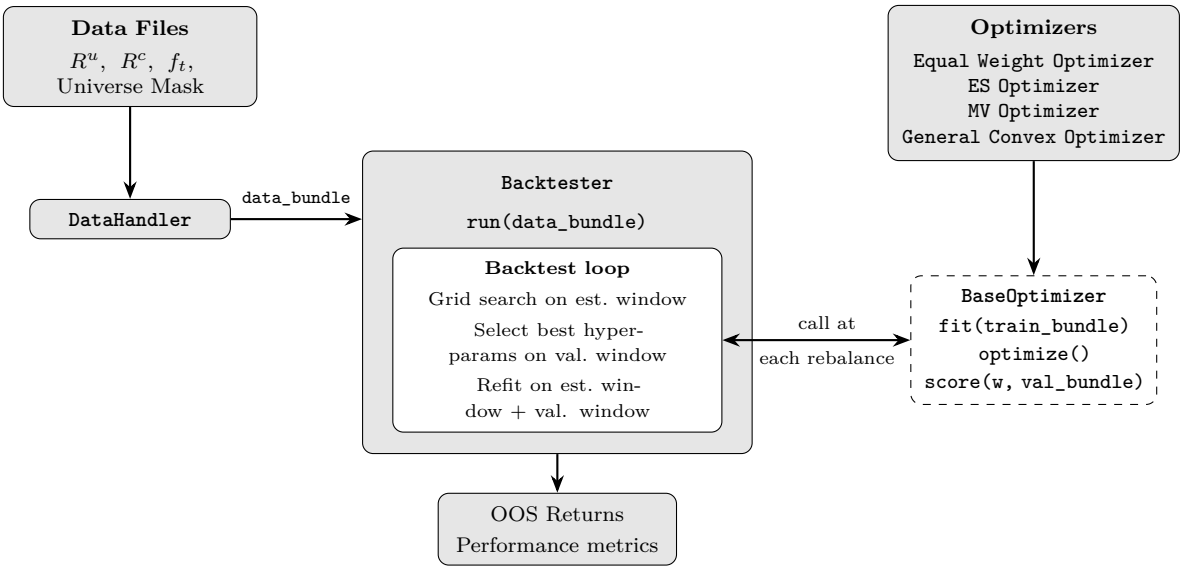


Figure 4.4: Schematic of the backtesting framework.

Data ingestion and alignment are implemetned in the `DataHandler`. It reads the raw inputs, the local unhedged asset returns R^u, the currency returns R^c, the forward premia f_t, and the universe mask that identifies which assets constitute the investable universe at each date, and aligns them into a `data_bundle` which is passed to the backtester. The optimizers operate only on the bundled series, and the construction of returns from these inputs follows Section 3.1.

All implemented optimizers inherit from an abstract base class, `BaseOptimizer`, that defines the three-method interface through which the backtester runs them. The `fit` method receives a training data bundle and estimates and stores the quantities the optimizer requires for its objective (e.g. covariance matrix for the mean-variance optimizer), so that estimation is performed entirely within each optimizer. The `optimize` method solves the specific optimization problem. The `score` method takes a set of weights and a validation bundle and returns the validation performance used for hyperparameter selection. Concrete subclasses implement this interface for each strategy, an equal-weight optimizer that serves as the naive benchmark, the mean-variance and Expected Shortfall optimizers, and a general convex optimizer TODO: Update this when done implementing generalization. The estimators and solver internals invoked inside the methods `fit` and `optimize` are described in Section 4.3.

[2]The full codebase with the backtest implementation is publicly available on GitHub: https://github.com/anejrozman/Multi-Currency-Portfolio-Optimization-with-General-Risk-Measures.

The backtest itself is implemented in the `Backtester.run` method, which advances through the data sample on a set rebalancing frequency (daily, weekly, monthly, quarterly, or annual). The empirical analysis in Chapter 5 and Appendix **??** uses monthly and quarterly rebalancing on the daily data. At each rebalancing date, the optimizer is calibrated and deployed using a fixed rolling lookback window, partitioned into an estimation window and a validation window. Hyperparameters are selected by a grid search over this partition. Each configuration is fitted on the estimation window via `fit`, used to produce weights via `optimize`, and evaluated on the validation window via `score`, and the configuration with the best validation score is retrained on both the estimation and validation windows and finally used to produce the portfolio weights for the next out-of-sample period. These weights are held until the next rebalancing date, the window then rolls forward and the procedure repeats. Because we keep the estimation, validation, and total lookback window lengths fixed throughout the backtest (10 years split into 9 for estimation and 1 for validation) and across universe sizes, every strategy is evaluated on the same out-of-sample dataset.

Each invocation of the backtester for a given optimizer produces a self-contained run directory. A new `run_<n>` folder is created under `run/<optimizer_name>/` and collects the full record of that run. The out-of-sample return series `oos_returns.csv`, the performance metrics `metrics.json`, a cumulative-return plot `cum_returns_plot.png`, and a configuration file `config.json` recording the run specification. Storing the configuration alongside its outputs makes each result reproducible and attributable to the settings that generated it.

4.3 Numerical Implementation

The optimization problems derived in Chapter 3 are not assembled in their explicit matrix form in code. Each problem is instead expressed directly in the high-level modeling syntax of `CVXPY`, converted automatically into the standard form, and finally passed to the solver. For example, the L_1 penalty is written with `CVXPY`'s `norm1` atom, so the positive-negative variable split used in the derivation is internal to the solver and never appears explicitly in the code. The resulting problem class depends only on the regularization. The Expected Shortfall optimizer is a linear program when it is left unregularized, since both the Rockafellar–Uryasev objective and its constraints are linear, whereas the L_2 ridge penalty introduces a quadratic term that turns every regularized Expected Shortfall problem into a quadratic program. The mean–variance problems are quadratic programs by construction, through the variance term in their objective.

All programs are solved with Gurobi, a state-of-the-art solver for linear and quadratic programs, accessed through `CVXPY`.[3] The solver is configured to use the barrier (interior point) method with tight convergence and accuracy tolerances, so that the large number of problems solved across rebalancing dates, universes, and hyperparameter configurations return reproducible results. The exact configuration is

```
problem.solve(
    solver=cp.GUROBI,
    verbose=False,
    BarConvTol=1e-8,
    Method=2,
    ScaleFlag=0,
    AggFill=0,
    PreDual=1,
```

[3] Gurobi is a commercial solver and requires a license to use, a free fallback solver is available in the project codebase.

```
    Presolve=1,
    Threads=3,
    FeasibilityTol=1e-9,
    OptimalityTol=1e-9,
)
```

The barrier method (`Method=2`) is paired with a barrier convergence tolerance of 10^{-8} and feasibility and optimality tolerances of 10^{-9}, the remaining flags specify the presolve, scaling, and threading behavior, which were optimized for the problem sizes considered in the thesis.

Hyperparameter selection follows the walk-forward grid search described in Section 4.2. The tuned hyperparameters are the L_1 and L_2 regularization coefficients applied separately to the asset and currency weights, λ_1^a, λ_2^a, λ_1^c, and λ_2^c, and the grid is the full cartesian product of their individual ranges. Each candidate configuration is scored by computing, on the validation window, the objective the optimizer minimizes, evaluated at the weights that configuration produces. The other parameters specific to the optimziation problems (these include the risk-aversion coefficient, the leverage budget, and the net currency exposure bounds) are fixed a priori rather than tuned. For each penalty the lower bound is zero, corresponding to the absence of that regularization term, while the upper bounds differ across parameters and are chosen so that larger values leave the optimal weights essentially unchanged. The resulting grids are reported in Table 4.2. TODO: Update table

Hyperparameter	Description	Grid range
λ_1^a	Asset L_1 penalty	$[0,\ 5 \times 10^{-3}]$
λ_2^a	Asset L_2 penalty	$[0,\ 3]$
λ_1^c	Currency L_1 penalty	$[0,\ 5 \times 10^{-4}]$
λ_2^c	Currency L_2 penalty	$[0,\ 5 \times 10^{-5}]$

Table 4.2: Hyperparameter grids for the regularization penalties. Each parameter is searched over a range spanning from zero up to the reported maximum, with the positive values placed on a log-spaced grid.

4.4 Performance Evaluation Metrics

All portfolio optimization strategies are evaluated on a common set of performance metrics computed from their out-of-sample return series. Because each optimizer minimizes a different objective, the strategies cannot be ranked solely by their in-sample objective values and are instead judged on the same realized metrics, each capturing a different view of the performance. All metrics are computed on USD out-of-sample returns net of the transaction costs (Derivation detailed in Appendix C). Throughout this section, $R_{\mathcal{P},t}$ denotes the realized daily out-of-sample portfolio return, $\bar{R}_{\mathcal{P}}$ its sample mean, σ its sample standard deviation, and R_f the risk-free rate. Daily figures are annualized using the standard 252 trading days per year assumption. The reported metrics are the expected shortfall at the 80% and 95% confidence levels, annualized volatility, maximum drawdown, the Sharpe ratio, the Sortino ratio, the expected shortfall ratio, and asset and currency turnover.

Expected shortfall at confidence level β measures the average loss in the worst $(1-\beta)$ tail of the return distribution. It is preferred to value at risk because it is coherent [1] and, rather than reporting only the loss threshold, averages the losses beyond it, so it reflects how severe the tail outcomes are and not merely how often they occur.[4] For the expected shortfall strategies,

[4] Formal definitions of value at risk and expected shortfall at confidence level β are given in Appendix A.1.

it is the quantity the optimizer minimizes, which makes it the most direct out-of-sample test of whether the expected shortfall strategy delivers on its objective. Because every strategy, including the mean–variance and naive benchmarks, is evaluated on the same measure, it also reveals whether explicitly targeting expected shortfall minimization produces lower realized tail risk than controlling variance alone. We evaluate it at $\beta = 0.80$ (the same confidence level used in the optimizer) and $\beta = 0.95$, which captures a more extreme region of the loss distribution. Reporting both levels separates a strategy's realized tail risk at the quantile it was optimized for from its tail risk at a deeper quantile it was not, showing whether (how much) the downside protection generalizes beyond the optimization level.

Annualized volatility is the standard deviation of portfolio returns. It is the classical measure of risk in the portfolio optimization literature, but does have some limitations. It is symmetric, meaning it penalizes favorable and adverse deviations equally, which is the limitation that motivates the downside and tail-risk measures reported alongside it. For the mean–variance strategies it plays the role that expected shortfall plays for the expected shortfall strategies. It is the objective the mean–variance optimizers minimize, and so it is the most direct out-of-sample test of whether they deliver on their objective. Because a portfolio can achieve low volatility while still suffering severe cumulative losses over time, it is informative only when interpretedtogether with the maximum drawdown and other tail-risk measures.

Maximum drawdown is the largest decline of the cumulative return path over the out-of-sample period. Unlike expected shortfall and volatility, which summarize the one-period return distribution, drawdown is a property of the realized wealth path, capturing the cumulative and sequential experience of losses that an investor goes through. Such path-dependent losses matter in practice. For example, deep drawdowns can breach margin requirements. Importantly, none of the optimizers considered in the thesis targets drawdown directly, so it serves as an out-of-sample robustness check on a metric the strategies were not built to control. Letting $V_t = \prod_{s \le t}(1 + R_{\mathcal{P},s})$ denote the cumulative wealth of portfolio $\mathcal{P}$, with $V_0 = 1$, and $P_t = \max_{s \le t} V_s$ its running maximum, the maximum drawdown is defined as

$$\mathrm{MD} = \max_t \frac{P_t - V_t}{P_t}.$$

The Sharpe ratio is the annualized ratio of the average excess portfolio return relative to its total volatility. It is the standard measure of return per unit of total risk. The Sharpe ratio is given by

$$\mathrm{SR} = \frac{\bar{R}_{\mathcal{P}} - R_f}{\sigma}\sqrt{252}.$$

The Sortino ratio replaces volatility in the Sharpe ratio with downside volatility, so that a strategy is not penalized for upside dispersion. By measuring volatility only over adverse outcomes it addresses the main downside of the Sharpe ratio. Read against the Sharpe ratio, it indicates whether a strategy's risk-adjusted performance stems from downside protection or from lower overall variability. It is the annualized ratio of average excess return to the standard deviation of the negative returns σ_{down}, given by

$$\mathrm{SOR} = \frac{\bar{R}_{\mathcal{P}} - R_f}{\sigma_{\mathrm{down}}}\sqrt{252}.$$

The expected shortfall ratio is the tail-risk analogue of the Sharpe ratio, reporting average return per unit of downside risk [24]. Where the Sharpe ratio scales return by total volatility and the Sortino ratio by downside deviation, the expected shortfall ratio scales it by the tail loss itself. A strategy scores well only if it earns return efficiently relative to its realized tail

risk, so the ratio rewards the trade-off the expected shortfall optimizers are solving. Reported in annualized units alongside the Sharpe and Sortino ratios, the three can be read together to trace how the ranking of strategies shifts as the risk moves from total, to downside, to tail risk. It is computed from the annualized average return and the annualized expected shortfall

$$\text{ESR} = \frac{\bar{R}_{\mathcal{P}}}{\text{ES}_{\beta}}.$$

We use the expected shortfall ratio at the $\beta = 0.80$ confidencelevel.

Finally, turnover proxies the trading activity, and the implementation costs, of each strategy. Although the reported returns are already net of proportional transaction costs, turnover remains informative because it measures how much the portfolio must trade to maintain its positions, its capacity to scale, and its exposure to frictions such as slippage and market impact that our model does not capture. A strategy whose performance depends on heavy rebalancing can be hard to implement, so low and stable turnover is a desirable property. Turnover is reported separately for the asset and currency legs, since the two are traded through different instruments with different cost levels. Asset turnover is the average rebalancing volume over the out-of-sample period

$$\text{Turnover}^{a} = \frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{N}|x_{i,t} - x_{i,t}^{-}|,$$

where $x_{i,t}^{-}$ is the asset weight just before rebalancing at t and $x_{i,t}$ the weight set at the rebalance, and T is the number of out-of-sample rebalancing dates. Currency turnover is the average absolute notional of the forward positions

$$\text{Turnover}^{c} = \frac{1}{T}\sum_{t=1}^{T}\sum_{c=1}^{M}|\phi_{c,t}|.$$

Chapter 5

Empirical Analysis: The Joint–Overlay Crossover

In this chapter, we present an empirical analysis of the multi-currency portfolio optimization problem using historical data. We apply the model-free return representation derived in chapter **??** to compute realized returns for the portfolio strategies derived in section **??**. We evalutate the performance of the various strategies on two investment universes, a large investment universe consisting of 50 biggest stocks in the 14 largest developed markets and a small investment universe consisting of 10 biggest stocks in the same markets. We compare the performance of the optimized portfolios to a benchmark equally weighted portfolio.

5.1 Backtest Results of Expected Shortfall Optimization Approaches

We can see in table 5.1 that the performance of the expected shortfall optimization approaches varies across different universe sizes. In the small universe, the optimized portfolios generally outperform the equal weight benchmark in terms of risk-adjusted returns, as measured by the Sharpe Ratio and Sortino Ratio. However, as we increase the universe size to medium and large, the performance of the optimized portfolios relative to the equal weight benchmark deteriorates, with some strategies even underperforming the benchmark in terms of risk-adjusted returns. This suggests that the benefits of optimization may diminish as the investment universe expands, potentially due to increased estimation error and model uncertainty in larger universes.

Panel A: Small Investment Universe (top 10 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	1.15	0.71	0.98	0.69	0.67	0.67	0.66
Vol (%)	13.96	8.51	11.84	8.18	7.99	7.97	7.85
MD (%)	54.76	30.81	44.28	29.31	29.77	28.31	28.89
SR	0.12	0.11	0.20	0.08	0.13	0.08	0.17
SOR	0.17	0.13	0.27	0.10	0.18	0.11	0.23
ESR	0.09	0.08	0.15	0.06	0.10	0.06	0.13
Turnover^a	0.29	1.20	1.24	1.28	1.06	1.26	1.15
Turnover^c	0.92	0.93	0	2.39	1.67	2.98	1.77
Panel B: Medium Investment Universe (top 50 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	1.04	0.43	0.80	0.44	0.42	0.43	
Vol (%)	12.50	5.84	9.87	5.68	5.73	5.53	
MD (%)	55.74	38.54	42.37	38.59	38.56	38.84	
SR	0.32	0.56	0.45	0.56	0.55	0.54	
SOR	0.44	0.77	0.64	0.75	0.77	0.73	
ESR	0.24	0.48	0.36	0.45	0.48	0.43	
Turnover^a	0.27	1.60	1.58	1.78	1.52	1.78	
Turnover^c	0.92	0.95	0	0.97	1.26	1.67	
Panel C: Large Investment Universe (top 100 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	0.97	0.34	0.76	0.35	0.33	0.35	
Vol (%)	11.67	4.46	9.21	4.57	4.45	4.55	
MD (%)	56.26	31.60	35.12	31.32	30.61	30.45	
SR	0.37	0.89	0.57	0.72	0.87	0.76	
SOR	0.49	1.18	0.80	0.95	1.14	1.01	
ESR	0.28	0.76	0.44	0.59	0.74	0.62	
Turnover^a	0.27	1.61	1.61	1.89	1.42	1.91	
Turnover^c	0.91	0.96	0	0.97	1.15	1.43	

Table 5.1: Performance metrics for Expected Shortfall optimization approaches on increasing universe sizes

Table 5.1 reports out-of-sample performance metrics for the international portfolio optimization under expected shortfall, with portfolios rebalanced quarterly. The realized out-of-sample statistics shown are Expected Shortfall (ES), volatility (Vol), maximum drawdown (MD), the Sharpe Ratio (SR), the Sortino Ratio (SOR), the Expected Shortfall Ratio (ESR), and average asset and currency turnover, denoted Turnover^a and Turnover^c, respectively. Each panel corresponds to a different investment universe: Panel A covers the small universe, made up of the 10 largest companies by market capitalization in each economy; Panel B the medium universe, comprising the 50 largest companies per economy; and Panel C the large universe, consisting of the 100 largest companies per economy. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. Details on the optimization approaches and performance metrics are provided in section **??**.



Figure 5.1: Cumulative returns of expected shortfall optimization approaches in the small investment universe.

Figure 5.1 shows the cumulative returns of the expected shortfall optimization approaches in the small investment universe, compared to the equal weight benchmark. The portfolios are reported net of transaction costs, and the sample period runs from June 1990 to June 2023, with the first out-of-sample portfolio formed in June 2000. Rebalancing is performed quarterly.

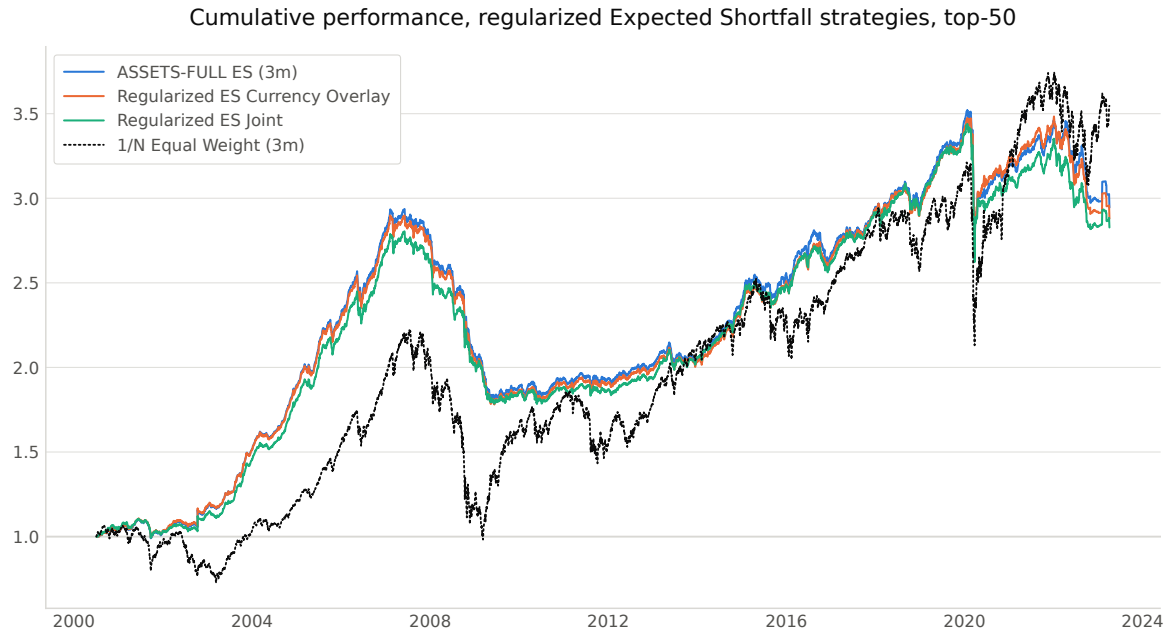


Figure 5.2: Cumulative returns of expected shortfall optimization approaches in the medium investment universe.

Figure 5.2 shows the cumulative returns of the expected shortfall optimization approaches in the medium investment universe, compared to the equal weight benchmark. The portfolios are reported net of transaction costs, and the sample period runs from June 1990 to June 2023, with the first out-of-sample portfolio formed in June 2000. Rebalancing is performed quarterly.

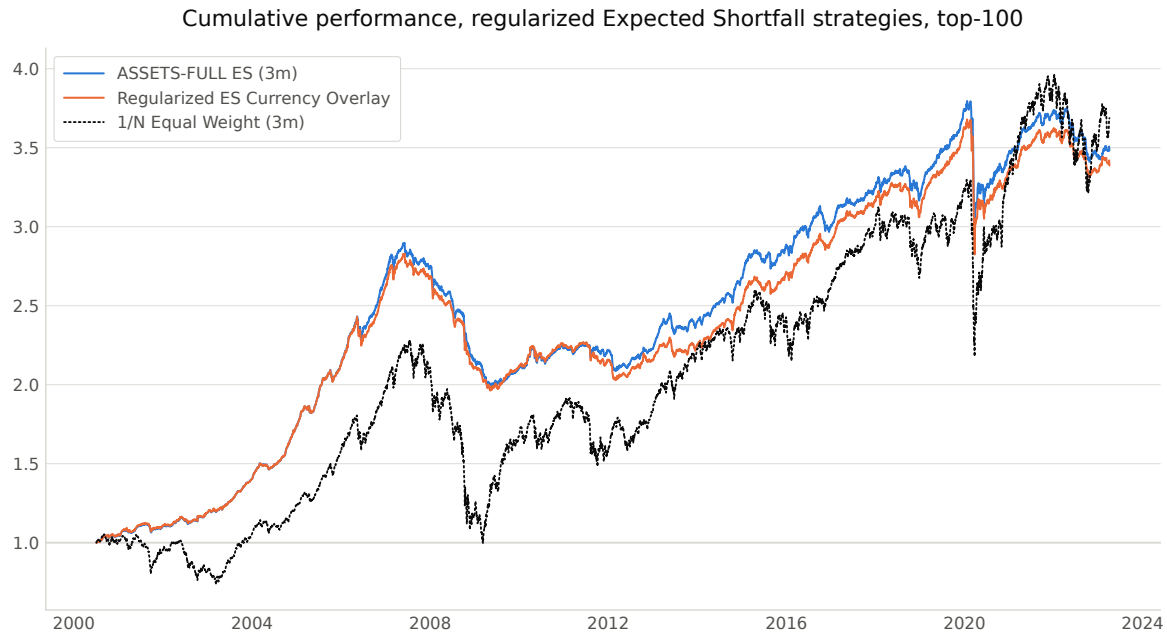


Figure 5.3: Cumulative returns of expected shortfall optimization approaches in the large investment universe.

Figure 5.3 shows the cumulative returns of the expected shortfall optimization approaches in the large investment universe, compared to the equal weight benchmark. The portfolios are reported net of transaction costs, and the sample period runs from June 1990 to June 2023, with the first out-of-sample portfolio formed in June 2000. Rebalancing is performed quarterly.

List of results interpretations to write up

- **Optimizers vs 1/N** All strategies outperform 1/N on all risk metrics and SR, SOR and ESR in all 3 universe sizes. The risk reduction is larger in ES and Vol than in MD, suggesting that the optimized portfolios are better at controlling tail risk and volatility, but may still be sucseptible to large drawdowns.
- **Risk metrics with increasing universe size** We can observe that the risk metrics for the 1/N portfolio do not improve by much as the universe size increases. They do increase substantially for the optimized portfolios. They drop by around 50% from the small to the large universe, while the 1/N portfolio only drops by around 15%. This suggests that the benefits of optimization are more pronounced in larger universes, where there is more opportunity to diversify and reduce risk.
- **Joint-Reg vs. Overlay-Reg** In the small universe, Joint-Reg is ahead on SR, SOR and ESR at slightly higher. Still waiting for results on the medium and large universes. Hypothesis: with the large universe there is no more benefit to the joint optimization.
- **Regularized vs. unregularized.** Overlay vs. Overlay-Reg and Joint vs. Joint-Reg. Regularization does not improve risk metrics but does improve ER, SOR and ESR. It typically reduces turnover, but not in every situation.
- **Plots** TODO: Compare drawdown depth of the optimizers against 1/N around 2002, 2008–09 and 2020, and compare how that gap changes based on universe size.

5.2 Backtest Results of Mean-Variance Optimization Approaches

Panel A: Small Investment Universe (top 10 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	1.15						
Vol (%)	13.96						
MD (%)	54.76						
SR	0.12						
SOR	0.17						
ESR	0.09						
Turnover^a	0.29						
Turnover^c	0.92						
Panel B: Medium Investment Universe (top 50 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	1.04						
Vol (%)	12.50						
MD (%)	55.74						
SR	0.32						
SOR	0.44						
ESR	0.24						
Turnover^a	0.27						
Turnover^c	0.92						
Panel C: Large Investment Universe (top 100 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	0.97						
Vol (%)	11.67						
MD (%)	56.26						
SR	0.37						
SOR	0.49						
ESR	0.28						
Turnover^a	0.27						
Turnover^c	0.91						

Table 5.2: Performance metrics for mean-variance optimization approaches on increasing universe sizes

Table 5.2 reports out-of-sample performance metrics for the international portfolio optimization under mean-variance, with portfolios rebalanced quarterly. The realized out-of-sample statistics shown are Expected Shortfall (ES), volatility (Vol), maximum drawdown (MD), the Sharpe Ratio (SR), the Sortino Ratio (SOR), the Expected Shortfall Ratio (ESR), and average asset and currency turnover, denoted Turnover^a and Turnover^c, respectively. Each panel corresponds to a different investment universe: Panel A covers the small universe, made up of the 10 largest companies by market capitalization in each economy; Panel B the medium universe, comprising the 50 largest companies per economy; and Panel C the large universe, consisting of the 100 largest companies per economy. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. Details on the optimization approaches and performance metrics are provided in section **??**.

5.3 The Effects of Universe Size on Portfolio Performance

Chapter 6

Convex Risk Measures

Chapter 7

Empirical Analysis: End To End Portfolio Optimization

Chapter 8

Conclusion

8.1 Findings

8.2 Limitations and Future Research

Appendices

Appendix A

Optimization Derivation: Expected Shortfall

In this appendix, we derive the optimization problems used in the main text, building up from the Expected Shortfall objective to the implementable quadratic programs solved in the empirical analysis. In Section A.1 we follow [22] to derive the equivalence between direct minimization of Expected Shortfall and an auxiliary formulation suitable for linear programming. Following [24], in Section A.2 we embed this equivalence in a mean-ES objective, by replacing the conditional expectation with its empirical counterpart, and linearizing the terms to assemble the base unregularized portfolio optimization problem as a linear program. In Section A.3 we introduce three regularization techniques, an L_1 penalty, an L_2 penalty, and a leverage constraint, giving us the base regularized quadratic program for portfolio optimization. In Section A.4 we extend the framework to the multi-currency setting under the separate (currency overlay) approach, deriving the asset and the currency quadratic programs. Finally, in Section A.5 we derive the joint quadratic program formulation, in which asset and currency weights are optimized simultaneously through a single Expected Shortfall term that captures the combined return distribution.

A.1 Derivation of the Auxiliary Expected Shortfall Formulation

We derive the equivalence between the direct minimization of Expected Shortfall and the auxiliary formulation proposed by Rockafellar and Uryasev [22] that allows for formulating the portfolio optimization as a linear program.

We assume that the loss distribution is continuous, meaning the loss cumulative distribution function has no jumps and the probability density function $f(l)$ exists. Under these assumptions, Value-at-Risk (VaR) is uniquely defined [20].

Define the loss random variable at time $t+1$, given portfolio weights $\boldsymbol{x}_t$, as $L(\boldsymbol{x}_t) = -\boldsymbol{x}_t^T \boldsymbol{R}_{t+1}$. Let $f(l)$ denote the probability density function of the loss $L(\boldsymbol{x}_t)$. The VaR at confidence level β is defined as the β-quantile of the loss distribution

$$\mathrm{VaR}_\beta(\boldsymbol{x}_t) = \min\{\alpha : \mathbb{P}(L(\boldsymbol{x}_t) \leq \alpha) \geq \beta\}.$$

The Expected Shortfall at confidence level β is defined as the conditional expectation of losses exceeding VaR_β

$$\mathrm{ES}_\beta(\boldsymbol{x}_t) = \mathbb{E}[L(\boldsymbol{x}_t) \mid L(\boldsymbol{x}_t) \geq \mathrm{VaR}_\beta(\boldsymbol{x}_t)] = \frac{1}{1-\beta} \int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} l f(l)\, dl.$$

We now consider the auxiliary function F_β defined as

$$F_\beta(\boldsymbol{x}_t, \alpha) = \alpha + \frac{1}{1-\beta}\mathbb{E}_t[(L(\boldsymbol{x}_t) - \alpha)^+].$$

We can rewrite the expectation term using the integral form

$$\mathbb{E}_t[(L(\boldsymbol{x}_t) - \alpha)^+] = \int_{-\infty}^{\infty} (l-\alpha)^+ f(l)\, dl = \int_{\alpha}^{\infty} (l-\alpha) f(l)\, dl.$$

Before minimizing $F_\beta(\boldsymbol{x}_t, \alpha)$ with respect to α, we verify convexity. Using Leibniz's rule, the first and second derivative of F_β with respect to α is

$$\begin{aligned}
\frac{\partial F_\beta}{\partial \alpha} &= 1 + \frac{1}{1-\beta}\frac{\partial}{\partial \alpha}\int_{\alpha}^{\infty} (l-\alpha) f(l)\, dl \\
&= 1 + \frac{1}{1-\beta}\left[-f(\alpha)\cdot 0 + \int_{\alpha}^{\infty} (-1) f(l)\, dl\right] \\
&= 1 - \frac{1}{1-\beta}\int_{\alpha}^{\infty} f(l)\, dl \\
\frac{\partial^2 F_\beta}{\partial \alpha^2} &= \frac{1}{1-\beta} f(\alpha).
\end{aligned}$$

Since the probability density function $f(\alpha) \geq 0$ and $\beta < 1$, the second derivative is non-negative, $\frac{\partial^2 F_\beta}{\partial \alpha^2} \geq 0$. Thus, $F_\beta(\boldsymbol{x}_t, \alpha)$ is convex in α, ensuring that the first-order condition is necessary and sufficient for a global minimum.

Differentiation of $F_\beta(\boldsymbol{x}_t, \alpha)$ with respect to α gives us the first-order condition for optimality

$$\begin{aligned}
\frac{\partial F_\beta}{\partial \alpha} &= 1 - \frac{1}{1-\beta}\int_{\alpha}^{\infty} f(l)\, dl \\
&= 1 - \frac{1}{1-\beta}\mathbb{P}(L(\boldsymbol{x}_t) \geq \alpha).
\end{aligned}$$

Setting the derivative to zero to find the optimal α^*

$$\begin{aligned}
1 - \frac{1}{1-\beta}\mathbb{P}(L(\boldsymbol{x}_t) \geq \alpha^*) &= 0 \\
\mathbb{P}(L(\boldsymbol{x}_t) \geq \alpha^*) &= 1-\beta \\
\mathbb{P}(L(\boldsymbol{x}_t) \leq \alpha^*) &= \beta.
\end{aligned}$$

This implies that the optimal α^* that minimizes the auxiliary function for a fixed $\boldsymbol{x}_t$ is exactly the Value-at-Risk at level β, so $\alpha^* = \mathrm{VaR}_\beta(\boldsymbol{x}_t)$.

Substituting $\alpha^* = \mathrm{VaR}_\beta(\boldsymbol{x}_t)$ back into $F_\beta(\boldsymbol{x}_t, \alpha)$ we obtain

$$\begin{aligned}
F_\beta(\boldsymbol{x}_t, \mathrm{VaR}_\beta(\boldsymbol{x}_t)) &= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} (l - \mathrm{VaR}_\beta(\boldsymbol{x}_t)) f(l)\, dl \\
&= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\left(\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} l f(l)\, dl - \mathrm{VaR}_\beta(\boldsymbol{x}_t)\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} f(l)\, dl\right).
\end{aligned}$$

Recognizing that $\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} f(l)\, dl = \mathbb{P}(L(\boldsymbol{x}_t) \geq \mathrm{VaR}_\beta(\boldsymbol{x}_t)) = 1-\beta$, we have

$$\begin{aligned}
F_\beta(\boldsymbol{x}_t, \mathrm{VaR}_\beta(\boldsymbol{x}_t)) &= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} l f(l)\, dl - \frac{1}{1-\beta}\mathrm{VaR}_\beta(\boldsymbol{x}_t)(1-\beta) \\
&= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \mathrm{ES}_\beta(\boldsymbol{x}_t) - \mathrm{VaR}_\beta(\boldsymbol{x}_t) \\
&= \mathrm{ES}_\beta(\boldsymbol{x}_t).
\end{aligned}$$

Thus, minimizing $\mathrm{ES}_\beta(\boldsymbol{x}_t)$ with respect to $\boldsymbol{x}_t$ is equivalent to minimizing $F_\beta(\boldsymbol{x}_t, \alpha)$ jointly with respect to $\boldsymbol{x}_t$ and α

$$\min_{\boldsymbol{x}_t} \mathrm{ES}_\beta(\boldsymbol{x}_t) = \min_{\boldsymbol{x}_t, \alpha} \left(\alpha + \frac{1}{1-\beta} \mathbb{E}_t[(-\boldsymbol{x}_t^T \boldsymbol{R}_{t+1} - \alpha)^+] \right). \tag{A.1}$$

Finally, we observe that $F_\beta(\boldsymbol{x}_t, \alpha)$ is jointly convex in $(\boldsymbol{x}_t, \alpha)$. The term α is linear (hence convex). The inner map $(\boldsymbol{x}_t, \alpha) \mapsto (-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)$ is affine. The outer function $z \mapsto z^+$ is convex and nondecreasing. The composition of a convex function with an affine map is convex [2]. Thus $(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+$ is jointly convex in $(\boldsymbol{x}_t, \alpha)$. Taking the expectation preserves convexity.

Since the objective function is convex, any local minimum is global, ensuring the optimization problem can be solved reliably with standard convex optimization methods.

A.2 Linear Program Representation of Portfolio Optimization

The equivalence in (A.1) reduces Expected Shortfall minimization to the minimization of the jointly convex auxiliary function $F_\beta(\boldsymbol{x}_t, \alpha)$

$$\min_{\boldsymbol{x}_t} \mathrm{ES}_\beta(\boldsymbol{x}_t) = \min_{\boldsymbol{x}_t, \alpha} \left(\alpha + \frac{1}{1-\beta} \mathbb{E}_t[(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+] \right).$$

This formulation is not yet appropriate for portfolio optimization, as it only optimizes for downside risk, without taking into account the expected return of the portfolio. It also depends on the unknown expectation $\mathbb{E}_t[\cdot]$. We address this by incorporating the expected return into the optimization objective and replacing the expectation with its empirical estimator.

Following [7] and [24], we trade off expected return against downside risk through a relative risk-aversion coefficient $\gamma > 0$

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ -\boldsymbol{\mu}_t^\top \boldsymbol{x}_t + \frac{\gamma}{2} \mathrm{ES}_\beta(\boldsymbol{x}_t) \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \tag{A.2}$$

where $\boldsymbol{\mu}_t$ denotes the asset return mean estimator and the budget constraint $\boldsymbol{x}_t^\top \mathbf{1}_N = 1$ enforces full investment. Substituting the auxiliary representation (A.1) into (A.2) gives

$$\boldsymbol{x}_t^*, \alpha^* = \arg\min_{\boldsymbol{x}_t, \alpha} \left\{ -\boldsymbol{\mu}_t^\top \boldsymbol{x}_t + \frac{\gamma}{2}\alpha + \frac{\gamma}{2(1-\beta)} \mathbb{E}_t[(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+] \right\}.$$

Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q \in \mathbb{R}^N$ denote q historical realizations of the joint return vector $\boldsymbol{R}_{t+1}$. We approximate the conditional expectation by the empirical average over this sample

$$\mathbb{E}_t[(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+] \approx \frac{1}{q} \sum_{k=1}^{q} (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+,$$

so that the portfolio optimization problem becomes

$$\boldsymbol{x}_t^*, \alpha^* = \arg\min_{\boldsymbol{x}_t, \alpha} \left\{ -\boldsymbol{\mu}_t^\top \boldsymbol{x}_t + \frac{\gamma}{2}\alpha + \frac{\gamma}{2q(1-\beta)} \sum_{k=1}^{q} (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+ \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1. \tag{A.3}$$

The summand $(-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$ is convex but non-linear. We linearize it by introducing, for each observation k, an auxiliary variable $u_k \geq 0$ subject to

$$u_k \geq -\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha \qquad \Longleftrightarrow \qquad \boldsymbol{x}_t^\top \boldsymbol{y}_k + \alpha + u_k \geq 0. \tag{A.4}$$

Since the objective in (A.3) is strictly increasing in each u_k (its coefficient $\frac{\gamma}{2q(1-\beta)} > 0$) and minimized, every u_k is driven to its lower bound at the optimum, so $u_k^* = (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$. The slack constraints therefore reproduce the term $(-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$ while keeping all relations linear.

By collecting the decision variables in the vector

$$\tilde{\boldsymbol{x}}_t = \begin{bmatrix} x_{1,t}, \dots, x_{N,t}, & \alpha, & u_1, \dots, u_q \end{bmatrix}^\top = \begin{bmatrix} \boldsymbol{x}_t^\top, & \alpha, & \boldsymbol{u}^\top \end{bmatrix}^\top \in \mathbb{R}^{(N+q+1)},$$

and constants in the vector

$$\boldsymbol{c} = \begin{bmatrix} -\boldsymbol{\mu}_t^\top, & \frac{\gamma}{2}, & \frac{\gamma}{2q(1-\beta)} \mathbf{1}_q^\top \end{bmatrix}^\top \in \mathbb{R}^{(N+q+1)},$$

the portfolio optimization problem (A.3) can be written as a linear program

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* = \arg\min_{\tilde{\boldsymbol{x}}_t} \quad & \boldsymbol{c}^\top \tilde{\boldsymbol{x}}_t \\ \text{s.t.} \quad & \boldsymbol{D}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_{q+1}^\top \end{bmatrix} \tilde{\boldsymbol{x}}_t = E = 1, \\ & \boldsymbol{A}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} -\boldsymbol{R}^{\text{data}} & -\mathbf{1}_q & -\boldsymbol{I}_q \end{bmatrix} \tilde{\boldsymbol{x}}_t \le \boldsymbol{b} = \mathbf{0}_q, \\ & \alpha \ge 0, \quad u_k \ge 0 \quad \text{for } k = 1, \dots, q. \end{aligned} \tag{A.5}$$

Here $\boldsymbol{D} \in \mathbb{R}^{1\times(N+q+1)}$ encodes the budget constraint, while $\boldsymbol{A} \in \mathbb{R}^{q\times(N+q+1)}$ and $\boldsymbol{b} \in \mathbb{R}^q$ encode the q inequality constraints arising from the linearization (A.4). The matrix $\boldsymbol{R}^{\text{data}} \in \mathbb{R}^{q\times N}$ is comprised of sampled return vectors $\boldsymbol{y}_k^\top$ row by row, $\mathbf{0}_q$ is the q-dimensional zero vector, and $\boldsymbol{I}_q$ the q-dimensional identity matrix.

Problem (A.5) has a linear objective subject to one linear equality and q linear inequalities, and is thus a linear program solvable with standard methods. This formulation serves as the base unregularized portfolio optimization problem under Expected Shortfall. Regularization terms and additional constraints are added in Section A.3 of the appendix.

A.3 Regularization

The base problem (A.5) is known to be numerically unstable in high-dimensional settings, producing extreme and concentrated weights [5, 10]. We stabilize it with three regularization techniques. An L_1-norm penalty that promotes sparsity, an L_2-norm penalty that promotes shrinkage, and a leverage constraint that bounds gross exposure. Each is appended to (A.5) while preserving its linear or quadratic structure. The L_1 penalty and the leverage constraint share a common variable decomposition, which we introduce first.

A.3.1 L_1 Regularization

The L_1 penalty $\lambda_1 \|\boldsymbol{x}_t\|_1 = \lambda_1 \sum_{i=1}^N |x_{i,t}|$, with $\lambda_1 \ge 0$, is non-linear through the absolute value. Following [23], we linearize it by splitting each weight into its positive and negative parts via the auxiliary variables $\Delta_{i,t}^+ \ge 0$ and $\Delta_{i,t}^- \ge 0$

$$x_{i,t} = \Delta_{i,t}^+ - \Delta_{i,t}^-, \qquad |x_{i,t}| = \Delta_{i,t}^+ + \Delta_{i,t}^-. \tag{A.6}$$

The second identity holds at the optimum. Minimizing the L_1 penalty drives the parts to their smallest feasible values, so at most one of $\Delta_{i,t}^+$, $\Delta_{i,t}^-$ is non-zero, giving $\Delta_{i,t}^+ = \max(x_{i,t}, 0)$ and $\Delta_{i,t}^- = \max(-x_{i,t}, 0)$. The L_1 penalty is then linear in the auxiliary variables

$$\lambda_1 \|\boldsymbol{x}_t\|_1 = \lambda_1 \mathbf{1}_N^\top (\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-).$$

We extend the decision vector to carry the two parts

$$\tilde{\boldsymbol{x}}_t = \left[\boldsymbol{x}_t^\top,\ \ \alpha,\ \ \boldsymbol{u}^\top,\ \ (\boldsymbol{\Delta}_t^-)^\top,\ \ (\boldsymbol{\Delta}_t^+)^\top\right]^\top \in \mathbb{R}^{3N+q+1},$$

and append the penalty coefficients to the cost vector

$$\tilde{\boldsymbol{c}} = \left[-\boldsymbol{\mu}_t^\top,\ \ \tfrac{\gamma}{2},\ \ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1\mathbf{1}_N^\top,\ \ \lambda_1\mathbf{1}_N^\top\right]^\top.$$

The decomposition (A.6) is enforced as N additional linear equality constraints, $\boldsymbol{x}_t+\boldsymbol{\Delta}_t^- -\boldsymbol{\Delta}_t^+ = \mathbf{0}_N$, alongside the original budget constraint. All relations remain linear, so the optimization problem is still a linear program after the addition of the L_1 penalty.

A.3.2 L_2 Regularization

The L_2 penalty $\lambda_2\|\boldsymbol{x}_t\|_2^2 = \lambda_2\boldsymbol{x}_t^\top\boldsymbol{x}_t$, with $\lambda_2 \geq 0$, is quadratic in the weights and is written as a quadratic form

$$\lambda_2\|\boldsymbol{x}_t\|_2^2 = \tfrac{1}{2}\,\boldsymbol{x}_t^\top\boldsymbol{Q}\,\boldsymbol{x}_t, \qquad \boldsymbol{Q} = 2\lambda_2\boldsymbol{I}_N.$$

Embedding this in the augmented vector $\tilde{\boldsymbol{x}}_t$, the penalty becomes $\frac{1}{2}\,\tilde{\boldsymbol{x}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{x}}_t$ with $\tilde{\boldsymbol{Q}}$ placing the block $2\lambda_2\boldsymbol{I}_N$ on the asset weights and zeros on the auxiliary variables

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} 2\lambda_2\boldsymbol{I}_N & \mathbf{0}_{N\times(2N+q+1)} \\ \mathbf{0}_{(2N+q+1)\times N} & \mathbf{0}_{(2N+q+1)\times(2N+q+1)} \end{bmatrix}.$$

Adding this term turns the L_1-augmented linear objective of the problem (A.5) into a quadratic objective $\tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{x}}_t + \frac{1}{2}\,\tilde{\boldsymbol{x}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{x}}_t$. Since $\tilde{\boldsymbol{Q}}$ is positive semi-definite, the objective remains convex, and the optimization problem becomes a quadratic program with linear constraints.

A.3.3 Leverage Constraint

The leverage constraint $\|\boldsymbol{x}_t\|_1 \leq \ell$, with $\ell \geq 1$, bounds the portfolio's gross exposure and guarantees a bounded feasible region even when $\lambda_1 = 0$. Reusing the decomposition (A.6), it is expressed as a single linear inequality

$$\|\boldsymbol{x}_t\|_1 = \mathbf{1}_N^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) \leq \ell. \tag{A.7}$$

The choice $\ell = 1$ rules out leverage entirely and is equivalent to a hard L_1 constraint. As a linear inequality, it preserves the quadratic program structure.

A.3.4 Quadratic Program Representation of the Regularized Portfolio Optimization

Adding all three regularization terms to the base problem (A.5), makes the portfolio optimization problem under Expected Shortfall a quadratic program with linear constraints

$$\begin{aligned}
\tilde{\boldsymbol{x}}_t^* &= \arg\min_{\tilde{\boldsymbol{x}}_t} \quad \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{x}}_t + \tfrac{1}{2}\,\tilde{\boldsymbol{x}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{x}}_t, \\
\text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_{q+1}^\top & \mathbf{0}_N^\top & \mathbf{0}_N^\top \\ \boldsymbol{I}_N & \mathbf{0}_{N\times(q+1)} & \boldsymbol{I}_N & -\boldsymbol{I}_N \end{bmatrix}\tilde{\boldsymbol{x}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_N \end{bmatrix}, \\
\tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} -\boldsymbol{R}^{\text{data}} & -\mathbf{1}_q & -\boldsymbol{I}_q & \mathbf{0}_{q\times N} & \mathbf{0}_{q\times N} \\ \mathbf{0}_N^\top & 0 & \mathbf{0}_q^\top & \mathbf{1}_N^\top & \mathbf{1}_N^\top \end{bmatrix}\tilde{\boldsymbol{x}}_t \leq \boldsymbol{b} = \begin{bmatrix} \mathbf{0}_q \\ \ell \end{bmatrix}, \\
\alpha \geq 0, &\quad u_k \geq 0, \quad \Delta_{i,t}^- \geq 0, \quad \Delta_{i,t}^+ \geq 0 \quad \text{for } i = 1,\ldots,N,\ k = 1,\ldots,q.
\end{aligned} \tag{A.8}$$

The first row of $\tilde{\boldsymbol{D}}$ enforces the budget constraint and the remaining N rows enforce the weight decomposition (A.6). The first q rows of $\tilde{\boldsymbol{A}}$ encode the inequality constraints from the linearization (A.4), and are extended with zeros for the new variables, and the final row of $\tilde{\boldsymbol{A}}$ imposes the leverage constraint (A.7).

A.4 Separate (Currency Overlay) Optimization

In the separate (currency overlay) portfolio optimization approach, the asset and currency allocations are optimized in two sequential steps. First, the asset weights $\boldsymbol{x}_t \in \mathbb{R}^N$ are optimized on the fully hedged returns, then, conditional on the optimal $\boldsymbol{x}_t^*$, the currency exposures $\boldsymbol{\psi}_t \in \mathbb{R}^M$ are optimized. Let

$$\begin{aligned}\boldsymbol{\mu}^{fh} &= \mathbb{E}_t[\boldsymbol{R}_{t+1}^{fh}] \in \mathbb{R}^N,\\ \boldsymbol{\mu}^{c} &= \mathbb{E}_t[\boldsymbol{e}_{t+1} - \boldsymbol{f}_t] \in \mathbb{R}^M,\end{aligned}$$

denote the expected fully hedged asset return and the expected excess currency return respectively. The two otimization steps are then given by

$$\begin{aligned}\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \Big\{ &-\boldsymbol{x}_t^\top \boldsymbol{\mu}^{fh} + \tfrac{\gamma}{2}\mathrm{ES}_\beta\left(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh}\right) + \lambda_1^a\|\boldsymbol{x}_t\|_1 + \lambda_2^a\|\boldsymbol{x}_t\|_2^2\Big\}\\ &\text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell,\end{aligned} \tag{A.9}$$

$$\begin{aligned}\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{\psi}_t} \Big\{ &-\boldsymbol{\psi}_t^\top \boldsymbol{\mu}^{c} + \tfrac{\gamma}{2}\,\mathrm{ES}_\beta\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)\Big)\\ &+ \lambda_1^c\|\boldsymbol{\psi}_t\|_1 + \lambda_2^c\|\boldsymbol{\psi}_t\|_2^2\Big\}.\end{aligned} \tag{A.10}$$

The regularization parameters are kept separate across assets $(\lambda_1^a, \lambda_2^a)$ and currencies $(\lambda_1^c, \lambda_2^c)$.

A.4.1 Asset Optimization

The asset optimization step (A.9) is identical to the regularized single-currency program (A.8). It carries a budget constraint, L_1 and L_2 penalty terms, and a leverage constraint. It is therefore obtained directly from (A.8) by substituting the matrix of historically sampled fully hedged returns $\boldsymbol{R}^{fh} \in \mathbb{R}^{q\times N}$ for $\boldsymbol{R}^{\text{data}}$, the mean $\boldsymbol{\mu}^{fh}$ for $\boldsymbol{\mu}_t$, and the asset-specific penalties λ_1^a, λ_2^a. With the augmented vector

$$\tilde{\boldsymbol{x}}_t = \Big[\boldsymbol{x}_t^\top,\ \ \alpha_x,\ \ \boldsymbol{u}_x^\top,\ \ (\boldsymbol{\Delta}_{x,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{x,t}^+)^\top\Big]^\top \in \mathbb{R}^{3N+q+1},$$

the cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \Big[-(\boldsymbol{\mu}^{fh})^\top,\ \ \tfrac{\gamma}{2},\ \ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1^a\mathbf{1}_N^\top,\ \ \lambda_1^a\mathbf{1}_N^\top\Big]^\top,$$

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} 2\lambda_2^a \boldsymbol{I}_N & \mathbf{0}_{N\times(2N+q+1)} \\ \mathbf{0}_{(2N+q+1)\times N} & \mathbf{0}_{(2N+q+1)\times(2N+q+1)}\end{bmatrix},$$

and the equality, inequality, and sign constraints are exactly those of (A.8) with $\boldsymbol{R}^{fh}$ in place of $\boldsymbol{R}^{\text{data}}$. The step yields the optimal asset weights $\boldsymbol{x}_t^*$, which are then fixed in the currency overlay optimization.

A.4.2 Currency Overlay Optimization

Conditional on $\boldsymbol{x}_t^*$, the currency step (A.10) minimizes the Expected Shortfall of the combined return $\boldsymbol{x}_t^{*\top}\boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)$ over the currency exposures $\boldsymbol{\psi}_t$. Applying the same auxiliary representation and the empirical approximation as in Section A.2, we introduce the auxiliary variable α_ψ and slack variables $u_{k,\psi} \geq 0$. Since $\boldsymbol{x}_t^*$ is now fixed, the slack constraint becomes

$$u_{k,\psi} \geq -\boldsymbol{\psi}_t^\top(\boldsymbol{e}_k - \boldsymbol{f}_t) - \boldsymbol{x}_t^{*\top}\boldsymbol{R}_k^{fh} - \alpha_\psi \quad \Longleftrightarrow \quad -\boldsymbol{\psi}_t^\top(\boldsymbol{e}_k - \boldsymbol{f}_t) - \alpha_\psi - u_{k,\psi} \leq \boldsymbol{x}_t^{*\top}\boldsymbol{R}_k^{fh},$$

where $\boldsymbol{R}_k^{fh}$ is the k-th sampled fully hedged return vector and $\boldsymbol{e}_k$ the k-th sampled currency return vector. The fixed asset-portfolio returns therefore appear on the right-hand side. We stack the sampled excess currency returns $(\boldsymbol{e}_k - \boldsymbol{f}_t)^\top$ row by row into $\boldsymbol{R}^c \in \mathbb{R}^{q\times M}$, and collect the fixed asset-portfolio returns in $\boldsymbol{R}_P^{fh} = \boldsymbol{R}^{fh}\boldsymbol{x}_t^* \in \mathbb{R}^q$. We linearize the currency L_1 penalty with the decomposition $\psi_{c,t} = \Delta_{\psi,c,t}^+ - \Delta_{\psi,c,t}^-$ as in Section A.3.1, and form the augmented vector

$$\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{\psi}_t^\top,\ \ \alpha_\psi,\ \ \boldsymbol{u}_\psi^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^+)^\top\right]^\top \in \mathbb{R}^{3M+q+1},$$

The cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \left[-(\boldsymbol{\mu}^c)^\top,\ \ \tfrac{\gamma}{2},\ \ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1^c\mathbf{1}_M^\top,\ \ \lambda_1^c\mathbf{1}_M^\top\right]^\top,$$

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} 2\lambda_2^c\boldsymbol{I}_M & \mathbf{0}_{M\times(2M+q+1)} \\ \mathbf{0}_{(2M+q+1)\times M} & \mathbf{0}_{(2M+q+1)\times(2M+q+1)} \end{bmatrix}.$$

The currency overlay optimization is then the quadratic program

$$\begin{aligned}
\boldsymbol{\psi}_t^* = \arg\min_{\tilde{\boldsymbol{\psi}}_t} \quad & \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{\psi}}_t + \tfrac{1}{2}\,\tilde{\boldsymbol{\psi}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{\psi}}_t, \\
\text{s.t.} \quad & \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{I}_M \quad \mathbf{0}_{M\times(q+1)} \quad \boldsymbol{I}_M \quad -\boldsymbol{I}_M\right]\tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{E}} = \mathbf{0}_M, \\
& \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\psi}}_t = \left[-\boldsymbol{R}^c \quad -\mathbf{1}_q \quad -\boldsymbol{I}_q \quad \mathbf{0}_{q\times M} \quad \mathbf{0}_{q\times M}\right]\tilde{\boldsymbol{\psi}}_t \leq \boldsymbol{b} = \boldsymbol{R}_P^{fh}, \\
& \alpha_\psi \geq 0, \quad u_{k,\psi} \geq 0, \quad \Delta_{\psi,c,t}^- \geq 0, \quad \Delta_{\psi,c,t}^+ \geq 0 \quad \text{for } c = 1,\dots,M,\ k = 1,\dots,q.
\end{aligned} \tag{A.11}$$

The currency optimization differs from the asset optimization (A.9) in the following ways. First, there is no budget constraint, the currency portfolio is a zero-investment portfolio, so the domestic exposure $\psi_{0,t}$ is inferred residually and the equality block $\tilde{\boldsymbol{D}}$ enforces only the weight decomposition, with right-hand side $\tilde{\boldsymbol{E}} = \mathbf{0}_M$ rather than $[1, \mathbf{0}_M^\top]^\top$. Second, the fixed asset leg shifts the inequality right-hand side from $\mathbf{0}_q$ to $\boldsymbol{R}_P^{fh} = \boldsymbol{R}^{fh}\boldsymbol{x}_t^*$. Third, no leverage constraint is imposed on the currency exposures through matrix $\tilde{\boldsymbol{A}}$. The L_1 and L_2 regularizing terms are as before, through the last two blocks of $\tilde{\boldsymbol{c}}$ and the main block of $\tilde{\boldsymbol{Q}}$, so the currency overlay remains a convex quadratic program.

A.5 Joint Optimization

In the joint portfolio optimization approach, asset and currency weights are optimized simultaneously rather than sequentially. We stack them into the combined decision vector $\boldsymbol{\theta}_t = [\boldsymbol{x}_t^\top, \boldsymbol{\psi}_t^\top]^\top \in \mathbb{R}^{N+M}$ and the combined return vector $\boldsymbol{R}_{t+1} = [\boldsymbol{R}_{t+1}^{fh\top}, (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)^\top]^\top \in \mathbb{R}^{N+M}$, with joint mean $\boldsymbol{\mu} = \mathbb{E}_t[\boldsymbol{R}_{t+1}] \in \mathbb{R}^{N+M}$. The joint optimization problem is

$$\begin{aligned}
\boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} \Big\{ & -\boldsymbol{\theta}_t^\top\boldsymbol{\mu} + \tfrac{\gamma}{2}\mathrm{ES}_\beta\left(\boldsymbol{\theta}_t^\top\boldsymbol{R}_{t+1}\right) + \|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 \Big\}, \\
& \text{s.t. } \boldsymbol{\theta}_t^\top\boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}\boldsymbol{\theta}_t\|_1 \leq \ell.
\end{aligned} \tag{A.12}$$

The following matrices and vector adapt the problem to the joint optimization setting

$$\boldsymbol{q}_{N,M} = \begin{bmatrix} \mathbf{1}_N \\ \mathbf{0}_M \end{bmatrix}, \quad \boldsymbol{L} = \begin{bmatrix} \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \mathbf{0}_{M\times M} \end{bmatrix},$$

$$\boldsymbol{S}_1 = \begin{bmatrix} \lambda_1^a \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \lambda_1^c \boldsymbol{I}_M \end{bmatrix}, \quad \boldsymbol{S}_2 = \begin{bmatrix} \sqrt{\lambda_2^a}\, \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \sqrt{\lambda_2^c}\, \boldsymbol{I}_M \end{bmatrix}.$$

The vector $\boldsymbol{q}_{N,M}$ applies the budget constraint to the asset weights only, so that $\boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = \boldsymbol{x}_t^\top \mathbf{1}_N = 1$, leaving the currency weights as a zero-investment overlay. The matrix $\boldsymbol{L}$ applies the leverage constraint to the asset weights only, giving $\|\boldsymbol{L}\boldsymbol{\theta}_t\|_1 = \|\boldsymbol{x}_t\|_1$. The diagonal regularization matrices $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ allow the asset and currency penalties to differ within a single objective,

$$\|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 = \lambda_1^a\|\boldsymbol{x}_t\|_1 + \lambda_1^c\|\boldsymbol{\psi}_t\|_1, \qquad \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 = \lambda_2^a\|\boldsymbol{x}_t\|_2^2 + \lambda_2^c\|\boldsymbol{\psi}_t\|_2^2.$$

Applying the auxiliary representation of Section A.1 and the empirical approximation of Section A.2 to the joint return $\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}$, we introduce the auxiliary variable α and slack variables $u_k \geq 0$ for $k = 1, \ldots, q$, and linearize the L_1 penalty by decomposing all $N + M$ components of $\boldsymbol{\theta}_t$ into positive and negative parts as in Section A.3.1. Collecting the asset and currency parts separately, we form the augmented vector

$$\tilde{\boldsymbol{\theta}}_t = \Big[\, \boldsymbol{x}_t^\top,\ \ \boldsymbol{\psi}_t^\top,\ \ \alpha,\ \ \boldsymbol{u}^\top,\ \ (\boldsymbol{\Delta}_{x,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{x,t}^+)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^+)^\top \,\Big]^\top \in \mathbb{R}^{3N+3M+q+1}.$$

Let $\boldsymbol{R}^{\text{all}} = [\, \boldsymbol{R}^{fh} \ \ \boldsymbol{R}^c \,] \in \mathbb{R}^{q\times(N+M)}$ denote the matrix of jointly sampled asset and currency returns. The cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \Big[\, -\boldsymbol{\mu}^\top,\ \ \tfrac{\gamma}{2},\ \ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1^a\mathbf{1}_N^\top,\ \ \lambda_1^c\mathbf{1}_M^\top,\ \ \lambda_1^a\mathbf{1}_N^\top,\ \ \lambda_1^c\mathbf{1}_M^\top \,\Big]^\top,$$

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} 2\lambda_2^a\boldsymbol{I}_N & \mathbf{0}_{N\times M} & \mathbf{0}_{N\times(2N+2M+q+1)} \\ \mathbf{0}_{M\times N} & 2\lambda_2^c\boldsymbol{I}_M & \mathbf{0}_{M\times(2N+2M+q+1)} \\ \mathbf{0}_{(2N+2M+q+1)\times N} & \mathbf{0}_{(2N+2M+q+1)\times M} & \mathbf{0}_{(2N+2M+q+1)\times(2N+2M+q+1)} \end{bmatrix}.$$

The joint portfolio optimization problem is then the quadratic program

$$\begin{aligned} \tilde{\boldsymbol{\theta}}_t^* = \arg\min_{\tilde{\boldsymbol{\theta}}_t} \quad & \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{\theta}}_t + \tfrac{1}{2}\,\tilde{\boldsymbol{\theta}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{\theta}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\theta}}_t = & \begin{bmatrix} \boldsymbol{q}_{N,M}^\top & \mathbf{0}_{q+1}^\top & \mathbf{0}_{N+M}^\top & \mathbf{0}_{N+M}^\top \\ \boldsymbol{I}_{N+M} & \mathbf{0}_{(N+M)\times(q+1)} & \boldsymbol{I}_{N+M} & -\boldsymbol{I}_{N+M} \end{bmatrix}\tilde{\boldsymbol{\theta}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_{N+M} \end{bmatrix}, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\theta}}_t = & \begin{bmatrix} -\boldsymbol{R}^{\text{all}} & -\mathbf{1}_q & -\boldsymbol{I}_q & \mathbf{0}_{q\times 2N} & \mathbf{0}_{q\times 2M} \\ \mathbf{0}_{N+M}^\top & \mathbf{0}_{q+1}^\top & \boldsymbol{q}_{N,M}^\top & & \boldsymbol{q}_{N,M}^\top \end{bmatrix}\tilde{\boldsymbol{\theta}}_t \leq \boldsymbol{b} = \begin{bmatrix} \mathbf{0}_q \\ \ell \end{bmatrix}, \\ & \alpha \geq 0, \quad u_k \geq 0, \quad \text{for } k = 1, \ldots, q, \\ & \Delta_{x,i,t}^{\pm} \geq 0, \quad \Delta_{\psi,c,t}^{\pm} \geq 0, \quad \text{for } i = 1, \ldots, N,\ c = 1, \ldots, M. \end{aligned} \tag{A.13}$$

In $\tilde{\boldsymbol{D}}$, the first row enforces the asset budget constraint $\boldsymbol{x}_t^\top\mathbf{1}_N = 1$, and the remaining $N + M$ rows enforce the weight decomposition (A.6) for all components of $\boldsymbol{\theta}_t$. In $\tilde{\boldsymbol{A}}$, the first q rows encode the linearization inequalities (A.4) applied to the joint return matrix $\boldsymbol{R}^{\text{all}}$. The final row imposes the asset-only leverage constraint (A.7), giving $\mathbf{1}_N^\top(\boldsymbol{\Delta}_{x,t}^+ + \boldsymbol{\Delta}_{x,t}^-) = \|\boldsymbol{x}_t\|_1 \leq \ell$.

The joint formulation solves a single quadratic program over $\tilde{\boldsymbol{\theta}}_t$, capturing the interaction between asset and currency exposures through the joint Expected Shortfall term and the

off-diagonal return covariation in $\boldsymbol{R}^{\text{all}}$. This comes at the cost of a higher-dimensional problem relative to the two smaller programs of the separate approach.

All quadratic programs derived in this appendix share the same convex structure and are solvable with the same quadratic programming solver. The auxiliary representation of Expected Shortfall ensures this uniformity. Each extension of the program, from the base regularized single currency formulation (A.8), to the joint multi-currency program above (A.13), modifies only the return matrix and the constraint blocks that encode the budget, leverage, and weight decomposition, without changing the program structure.

Appendix B

Optimization Derivation: Mean-Variance

In this appendix, we derive the optimization problems used in the main text for mean-variance portfolio optimization. We start with the standard mean-variance formulation and then introduce regularization techniques and estimators to stabilize the optimization. We then extend the formulation to the multi-currency setting, considering separate optimization for assets and currency overlay, and finally the joint optimization problem. In Section B.1, we follow [25, 15] and present the quadratic program representation of the mean-variance portfolio optimization problem.

B.1 Quadratic Program Representation of the Mean-Variance Portfolio Optimization

The mean-variance portfolio optimization of [15] trades off the expected portfolio return against its return variance. The portfolio return $\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}$ is linear in the asset weights, so its variance is the quadratic form $\mathrm{Var}(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) = \boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t$, where $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$ is the asset return vector and $\boldsymbol{\Sigma}_t$ its covariance matrix. Following the international mean-variance formulation of [25], we write the portfolio optimization problem as a minimization that trades off expected return against variance through a relative risk-aversion coefficient $\gamma > 0$

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ -\boldsymbol{\mu}_t^\top \boldsymbol{x}_t + \frac{\gamma}{2} \boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \tag{B.1}$$

where $\boldsymbol{\mu}_t$ denotes the asset return mean estimator, $\boldsymbol{\Sigma}_t$ the asset return covariance estimator, and the budget constraint $\boldsymbol{x}_t^\top \mathbf{1}_N = 1$ enforces full investment.

In contrast to the Expected Shortfall objective (A.2), which depends on the unknown conditional expectation $\mathbb{E}_t[\cdot]$ and requires the auxiliary representation of Section A.1 together with a sample approximation to obtain a tractable form, the variance term in (B.1) is already expressed in closed quadratic form. Problem (B.1) is therefore already formulated as a quadratic program.

The objective in (B.1) is convex. The first term $-\boldsymbol{\mu}_t^\top \boldsymbol{x}_t$ is linear, hence convex, and the quadratic term $\frac{\gamma}{2} \boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t$ has Hessian $\gamma \boldsymbol{\Sigma}_t$. Since a covariance matrix is positive semi-definite and $\gamma > 0$, this Hessian is positive semi-definite, so the quadratic term, and with it the full objective, is convex [2]. Any local minimum is therefore global, ensuring the problem can be solved reliably with standard convex optimization methods.

The mean and covariance terms are unknown and replaced by their sample estimators $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$. Because the variance term is expressed in closed form, this step introduces no auxiliary

variables and leaves the decision vector at dimension N, independent of the number of historical observations q used to estimate it. This is different from the empirical approximation of the Expected Shortfall objective in Section A.2, where the conditional expectation is replaced by an average over q historical return samples and q slack variables u_k enlarge the problem to dimension $N + q + 1$. The sample covariance estimator we use is positive semi-definite, so the convexity established above is preserved under estimation. The choice of estimators is discussed in Section B.2.2.

Collecting the asset weights in the decision vector $\boldsymbol{x}_t \in \mathbb{R}^N$ and defining the cost vector and the penalty matrix

$$\boldsymbol{c} = -\boldsymbol{\mu}_t \in \mathbb{R}^N, \qquad \boldsymbol{Q} = \gamma \boldsymbol{\Sigma}_t \in \mathbb{R}^{N \times N},$$

the mean-variance problem (B.1) is written as the quadratic program

$$\begin{aligned} \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \quad & \boldsymbol{c}^\top \boldsymbol{x}_t + \tfrac{1}{2}\, \boldsymbol{x}_t^\top \boldsymbol{Q}\, \boldsymbol{x}_t, \\ \text{s.t.} \quad & \boldsymbol{D}\boldsymbol{x}_t = \mathbf{1}_N^\top \boldsymbol{x}_t = E = 1. \end{aligned} \tag{B.2}$$

Here $\boldsymbol{D} = \mathbf{1}_N^\top \in \mathbb{R}^{1 \times N}$ encodes the budget constraint with right-hand side $E = 1$.

Problem (B.2) has a convex quadratic objective subject to a single linear equality constraint, and is thus a quadratic program solvable with standard methods. This formulation serves as the base unregularized portfolio optimization problem under the mean-variance criterion. Regularization terms and additional constraints are added in Section B.2.

B.2 Regularization

The base problem (B.2) is well-documented to suffer from instability. The weights are sensitive to estimation error in $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$ and tend to take extreme, concentrated values, so that portfolios formed from the raw sample moments perform poorly out of sample [18, 25]. We stabilize the problem with two complementary approaches. First, we regularize the weights directly through an L_1-norm penalty that promotes sparsity, an L_2-norm penalty that promotes shrinkage, and a leverage constraint that bounds gross exposure. These are the same three techniques appended to the Expected Shortfall optimizationprogram in Section A.3. Second, we shrink the input estimators $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$ themselves toward more stable targets. Both preserve the convex quadratic structure of (B.2).

B.2.1 L_1, L_2 Regularization and Leverage Constraint

The L_1 penalty $\lambda_1 \|\boldsymbol{x}_t\|_1 = \lambda_1 \sum_{i=1}^N |x_{i,t}|$, with $\lambda_1 \geq 0$, is linearized just like in Appendix A.3.1. Each weight is split into its positive and negative parts through the auxiliary variables $\Delta_{i,t}^+ \geq 0$ and $\Delta_{i,t}^- \geq 0$, giving

$$\lambda_1 \|\boldsymbol{x}_t\|_1 = \lambda_1 \mathbf{1}_N^\top (\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-).$$

The leverage constraint $\|\boldsymbol{x}_t\|_1 \leq \ell$, with $\ell \geq 1$, reuses the same decomposition and is the single linear inequality

$$\|\boldsymbol{x}_t\|_1 = \mathbf{1}_N^\top (\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) \leq \ell.$$

As before, the choice $\ell = 1$ rules out leverage entirely and is equivalent to a hard L_1 constraint.

The L_2 penalty $\lambda_2 \|\boldsymbol{x}_t\|_2^2 = \lambda_2 \boldsymbol{x}_t^\top \boldsymbol{x}_t$, with $\lambda_2 \geq 0$, is the quadratic form $\frac{1}{2}\boldsymbol{x}_t^\top (2\lambda_2 \boldsymbol{I}_N)\boldsymbol{x}_t$, as in Appendix A.3.2. Here the mean-variance and Expected Shortfall cases differ. In the Expected Shortfall program, the L_2 term is what turns the linear program (A.5) into a quadratic program. The mean-variance objective (B.1) is already quadratic through the variance term,

so the L_2 penalty does not change the program class. Instead, it adds to the existing quadratic form and combines with the covariance into the single term

$$\tfrac{\gamma}{2}\,\boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t + \lambda_2 \boldsymbol{x}_t^\top \boldsymbol{x}_t = \tfrac{1}{2}\,\boldsymbol{x}_t^\top \left(\gamma \boldsymbol{\Sigma}_t + 2\lambda_2 \boldsymbol{I}_N\right) \boldsymbol{x}_t.$$

Because $\gamma\boldsymbol{\Sigma}_t$ is positive semi-definite and $2\lambda_2\boldsymbol{I}_N$ is positive definite for $\lambda_2 > 0$, the combined matrix is positive definite, so the penalized objective is strictly convex and admits a unique minimizer even when the covariance estimator $\boldsymbol{\Sigma}_t$ is singular, which occurs when the number of assets exceeds the number of observations. The combined term also makes the stabilizing role of the penalty explicit, since $\gamma\boldsymbol{\Sigma}_t + 2\lambda_2\boldsymbol{I}_N = \gamma(\boldsymbol{\Sigma}_t + \frac{2\lambda_2}{\gamma}\boldsymbol{I}_N)$ shrinks the covariance toward a scaled identity. The L_2 penalty can therefore be interpreted as a covariance shrinkage toward the identity [13], complementing the estimator shrinkage introduced in Section B.2.2.

B.2.2 Estimators

The program (B.2) depends on the unknown mean $\boldsymbol{\mu}_t$ and covariance $\boldsymbol{\Sigma}_t$. Replacing them by the sample mean $\bar{\boldsymbol{\mu}}_t$ and the sample covariance $\boldsymbol{S}_t$ is the approach, which is known to produce unstable and poorly diversified portfolios, as the optimization amplifies estimation error in these inputs [18, 11]. We instead replace each input by a shrinkage estimator, $\boldsymbol{\mu}_t = \boldsymbol{\mu}_t^{BS}$ and $\boldsymbol{\Sigma}_t = \boldsymbol{\Sigma}_t^{LW}$, following [25].

For estimating the mean, we use the Bayes–Stein estimator introduced in [9], which shrinks the sample mean towards a fixed shrinkage target. The shrinkage target is the return of the minimum-variance portfolio

$$\mu_0 = \frac{\mathbf{1}_N^\top \boldsymbol{S}_t^{-1} \bar{\boldsymbol{\mu}}_t}{\mathbf{1}_N^\top \boldsymbol{S}_t^{-1} \mathbf{1}_N},$$

and the Bayes–Stein estimator is the convex combination

$$\boldsymbol{\mu}_t^{BS} = (1-\hat{\omega})\,\bar{\boldsymbol{\mu}}_t + \hat{\omega}\,\mu_0\,\mathbf{1}_N, \qquad \hat{\omega} = \frac{N+2}{(N+2) + q\,(\bar{\boldsymbol{\mu}}_t - \mu_0 \mathbf{1}_N)^\top \boldsymbol{S}_t^{-1} (\bar{\boldsymbol{\mu}}_t - \mu_0 \mathbf{1}_N)},$$

where q is the number of observations in the estimation window, N the number of assets, and $\hat{\omega} \in [0,1]$ the shrinkage intensity. The intensity increases toward one as the sample size gets smaller, or as the variance of the sample means around μ_0 decreases. This pulls the estimator toward the shrinkage target when the data is least informative. The derivation of the estimator is given in [9].

For estimating the covariance matrix, we use the Ledoit–Wolf linear shrinkage estimator introduced in [12], which shrinks the sample covariance toward a shrinkage target. The Ledoit–Wolf estimator is the convex combination

$$\boldsymbol{\Sigma}_t^{LW} = \hat{v}\,\boldsymbol{F}_t + (1-\hat{v})\,\boldsymbol{S}_t,$$

with shrinkage intensity $\hat{v} \in [0,1]$. The shrinkage target $\boldsymbol{F}_t$ is the constant-correlation matrix, which retains the sample variances and replaces every pairwise correlation by the average sample correlation

$$F_{ii,t} = S_{ii,t}, \qquad F_{ij,t} = \bar{\rho}_t \sqrt{S_{ii,t}\,S_{jj,t}} \;\; (i \neq j), \qquad \bar{\rho}_t = \frac{2}{N(N-1)} \sum_{i<j} \frac{S_{ij,t}}{\sqrt{S_{ii,t}\,S_{jj,t}}}.$$

The intensity $\hat{v}$ is chosen to minimize the expected Frobenius distance between $\boldsymbol{\Sigma}_t^{LW}$ and the true covariance matrix, its closed form is derived in [12]. Both $\boldsymbol{F}_t$ and $\boldsymbol{S}_t$ are positive semi-definite, so $\boldsymbol{\Sigma}_t^{LW}$ is positive semi-definite and the convexity established in Section B.1 is preserved under estimation.

The estimators above are stated for a single asset block. In the multi-currency setting, the asset and currency means and covariances, and in the separate approach the covariance of the augmented currency-overlay problem, are constructed analogously, we defer their specification to Section B.3 and Section B.4.

B.2.3 Quadratic Program Representation of the Regularized Portfolio Optimization

Adding the three weight-level regularizers to the base problem (B.1) gives the regularized mean-variance problem

$$\begin{aligned} \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ -\boldsymbol{\mu}_t^{BS\top} \boldsymbol{x}_t + \tfrac{\gamma}{2}\, \boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t^{LW} \boldsymbol{x}_t + \lambda_1 \|\boldsymbol{x}_t\|_1 + \lambda_2 \|\boldsymbol{x}_t\|_2^2 \right\}, \\ \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell, \end{aligned} \tag{B.3}$$

Following the L_1 decomposition, we extend the decision vector to carry the positive and negative parts of the weights

$$\tilde{\boldsymbol{x}}_t = \left[\boldsymbol{x}_t^\top,\ \ (\boldsymbol{\Delta}_t^-)^\top,\ \ (\boldsymbol{\Delta}_t^+)^\top \right]^\top \in \mathbb{R}^{3N},$$

and define the cost vector and the penalty matrix

$$\tilde{\boldsymbol{c}} = \left[-\boldsymbol{\mu}_t^{BS\top},\ \ \lambda_1 \mathbf{1}_N^\top,\ \ \lambda_1 \mathbf{1}_N^\top \right]^\top, \qquad \tilde{\boldsymbol{Q}} = \begin{bmatrix} \gamma \boldsymbol{\Sigma}_t^{LW} + 2\lambda_2 \boldsymbol{I}_N & \mathbf{0}_{N\times 2N} \\ \mathbf{0}_{2N\times N} & \mathbf{0}_{2N\times 2N} \end{bmatrix}.$$

The regularized mean-variance portfolio optimization problem is then the quadratic program

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* &= \arg\min_{\tilde{\boldsymbol{x}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{x}}_t + \tfrac{1}{2}\, \tilde{\boldsymbol{x}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{x}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}} \tilde{\boldsymbol{x}}_t &= \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_N^\top & \mathbf{0}_N^\top \\ \boldsymbol{I}_N & \boldsymbol{I}_N & -\boldsymbol{I}_N \end{bmatrix} \tilde{\boldsymbol{x}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_N \end{bmatrix}, \\ \tilde{\boldsymbol{A}} \tilde{\boldsymbol{x}}_t &= \begin{bmatrix} \mathbf{0}_N^\top & \mathbf{1}_N^\top & \mathbf{1}_N^\top \end{bmatrix} \tilde{\boldsymbol{x}}_t \le b = \ell, \\ \Delta_{i,t}^- \ge 0, &\quad \Delta_{i,t}^+ \ge 0 \quad \text{for } i = 1, \dots, N. \end{aligned} \tag{B.4}$$

The first row of $\tilde{\boldsymbol{D}}$ enforces the budget constraint and the remaining N rows enforce the weight decomposition (A.6). The single row of $\tilde{\boldsymbol{A}}$ imposes the leverage constraint (A.7), giving $\mathbf{1}_N^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) = \|\boldsymbol{x}_t\|_1 \le \ell$.

Compared with the regularized Expected Shortfall program (A.8), the mean-variance program is smaller. Its decision vector has dimension $3N$ rather than $3N + q + 1$. This formulation serves as the regularized base problem under the mean-variance criterion. The asset and currency extensions under the separate and joint approaches are derived in Section B.3 and Section B.4.

B.3 Separate (Currency Overlay) Optimization

In the separate (currency overlay) portfolio optimization approach, the asset and currency allocations are optimized in two sequential steps. First, the asset weights $\boldsymbol{x}_t \in \mathbb{R}^N$ are optimized on the fully hedged returns, then, conditional on the optimal $\boldsymbol{x}_t^*$, the currency exposures $\boldsymbol{\psi}_t \in \mathbb{R}^M$ are optimized. Let

$$\begin{aligned} \boldsymbol{\mu}^{fh} &= \mathbb{E}_t[\boldsymbol{R}_{t+1}^{fh}] \in \mathbb{R}^N, & \boldsymbol{\Sigma}^a &= \mathrm{Var}_t(\boldsymbol{R}_{t+1}^{fh}) \in \mathbb{R}^{N\times N}, \\ \boldsymbol{\mu}^c &= \mathbb{E}_t[\boldsymbol{e}_{t+1} - \boldsymbol{f}_t] \in \mathbb{R}^M, & \boldsymbol{\Sigma}^c &= \mathrm{Var}_t(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \in \mathbb{R}^{M\times M}, \end{aligned}$$

denote the expected fully hedged asset return and excess currency return together with their covariance matrices, and let $\boldsymbol{\Sigma}^{ac} = \mathrm{Cov}_t(\boldsymbol{R}_{t+1}^{fh},\, \boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \in \mathbb{R}^{N\times M}$ denote the cross-covariance between fully hedged asset returns and excess currency returns. Each moment is the corresponding shrinkage estimator of Section B.2.2. The two optimization steps are then given by

$$\begin{aligned} \boldsymbol{x}_t^* = \underset{\boldsymbol{x}_t}{\arg\min} \Big\{ -\boldsymbol{x}_t^\top \boldsymbol{\mu}^{fh} + \tfrac{\gamma}{2}\, \boldsymbol{x}_t^\top \boldsymbol{\Sigma}^a \boldsymbol{x}_t + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \Big\} \\ \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell, \end{aligned} \tag{B.5}$$

$$\begin{aligned} \boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \underset{\boldsymbol{\psi}_t}{\arg\min} \Big\{ -\boldsymbol{\psi}_t^\top \boldsymbol{\mu}^c + \tfrac{\gamma}{2}\, \mathrm{Var}_t\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \Big) \\ + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \Big\}. \end{aligned} \tag{B.6}$$

The regularization parameters are kept separate across assets $(\lambda_1^a, \lambda_2^a)$ and currencies $(\lambda_1^c, \lambda_2^c)$.

B.3.1 Asset Optimization

The asset optimization step (B.5) is identical to the regularized single-currency program (B.4). It carries a budget constraint, L_1 and L_2 penalty terms, and a leverage constraint. It is therefore obtained directly from (B.4) by substituting the fully hedged asset moments $\boldsymbol{\mu}^{fh}$ for $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}^a$ for $\boldsymbol{\Sigma}_t$, together with the asset-specific penalties λ_1^a, λ_2^a. With the augmented vector

$$\tilde{\boldsymbol{x}}_t = \Big[\boldsymbol{x}_t^\top,\ \ (\boldsymbol{\Delta}_{x,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{x,t}^+)^\top \Big]^\top \in \mathbb{R}^{3N},$$

the cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \Big[-(\boldsymbol{\mu}^{fh})^\top,\ \ \lambda_1^a \mathbf{1}_N^\top,\ \ \lambda_1^a \mathbf{1}_N^\top \Big]^\top, \qquad \tilde{\boldsymbol{Q}} = \begin{bmatrix} \gamma \boldsymbol{\Sigma}^a + 2\lambda_2^a \boldsymbol{I}_N & \mathbf{0}_{N\times 2N} \\ \mathbf{0}_{2N\times N} & \mathbf{0}_{2N\times 2N} \end{bmatrix},$$

and the equality, inequality, and sign constraints are exactly those of (B.4) with $\boldsymbol{\Sigma}^a$ and $\boldsymbol{\mu}^{fh}$ in place of $\boldsymbol{\Sigma}_t$ and $\boldsymbol{\mu}_t$. The step yields the optimal asset weights $\boldsymbol{x}_t^*$, which are then fixed in the currency overlay optimization.

B.3.2 Currency Overlay Optimization

Conditional on $\boldsymbol{x}_t^*$, the currency step (B.6) optimizes the mean-variance objective of the combined hedged return $\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)$ over the currency exposures $\boldsymbol{\psi}_t$. Unlike the Expected Shortfall in the currency overlay of Section A.4.2, the variance decomposes in closed form

$$\mathrm{Var}_t\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \Big) = \boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}^a \boldsymbol{x}_t^* + 2\, \boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}^{ac} \boldsymbol{\psi}_t + \boldsymbol{\psi}_t^\top \boldsymbol{\Sigma}^c \boldsymbol{\psi}_t.$$

The term $\boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}^a \boldsymbol{x}_t^*$ does not depend on $\boldsymbol{\psi}_t$ and is therefore dropped from the minimization. Substituting the remaining terms into (B.6), the currency overlay objective becomes

$$\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \underset{\boldsymbol{\psi}_t}{\arg\min} \Big\{ -\boldsymbol{\psi}_t^\top \boldsymbol{\mu}^c + \tfrac{\gamma}{2}\, \boldsymbol{\psi}_t^\top \boldsymbol{\Sigma}^c \boldsymbol{\psi}_t + \gamma\, \boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}^{ac} \boldsymbol{\psi}_t + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \Big\}. \tag{B.7}$$

The fixed asset weights are present through the linear cross-covariance term $\gamma\, \boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}^{ac} \boldsymbol{\psi}_t$. Collecting the two linear terms shows that the coupling is equivalent to shifting the currency mean, $-\boldsymbol{\psi}_t^\top \boldsymbol{\mu}^c + \gamma\, \boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}^{ac} \boldsymbol{\psi}_t = -\boldsymbol{\psi}_t^\top (\boldsymbol{\mu}^c - \gamma\, (\boldsymbol{\Sigma}^{ac})^\top \boldsymbol{x}_t^*)$. The fixed asset portfolio therefore lowers the effective expected currency return by $\gamma\, (\boldsymbol{\Sigma}^{ac})^\top \boldsymbol{x}_t^*$. This effect is similar to the one

in Section A.4.2, where the fixed asset weights shift the right-hand side of the slack constraints to $\boldsymbol{R}_P^{fh} = \boldsymbol{R}^{fh}\boldsymbol{x}_t^*$.

We linearize the currency L_1 penalty with the decomposition $\psi_{c,t} = \Delta^+_{\psi,c,t} - \Delta^-_{\psi,c,t}$ as in Section A.3.1, and form the augmented vector

$$\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{\psi}_t^\top, \quad (\boldsymbol{\Delta}^-_{\psi,t})^\top, \quad (\boldsymbol{\Delta}^+_{\psi,t})^\top\right]^\top \in \mathbb{R}^{3M}.$$

The cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \left[-(\boldsymbol{\mu}^c)^\top + \gamma\, \boldsymbol{x}_t^{*\top}\boldsymbol{\Sigma}^{ac}, \quad \lambda_1^c \mathbf{1}_M^\top, \quad \lambda_1^c \mathbf{1}_M^\top\right]^\top, \qquad \tilde{\boldsymbol{Q}} = \begin{bmatrix} \gamma\boldsymbol{\Sigma}^c + 2\lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M\times 2M} \\ \mathbf{0}_{2M\times M} & \mathbf{0}_{2M\times 2M} \end{bmatrix}.$$

The currency overlay optimization is then the quadratic program

$$\begin{aligned} \tilde{\boldsymbol{\psi}}_t^* = \arg\min_{\tilde{\boldsymbol{\psi}}_t} \quad & \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\psi}}_t + \tfrac{1}{2}\, \tilde{\boldsymbol{\psi}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\psi}}_t, \\ \text{s.t.} \quad & \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{I}_M \quad \boldsymbol{I}_M \quad -\boldsymbol{I}_M\right] \tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{E}} = \mathbf{0}_M, \\ & \Delta^-_{\psi,c,t} \geq 0, \quad \Delta^+_{\psi,c,t} \geq 0 \quad \text{for } c = 1, \ldots, M. \end{aligned} \tag{B.8}$$

The currency optimization differs from the asset optimization (B.5) in the same ways as in the Expected Shortfall case. First, there is no budget constraint, as the currency portfolio is a zero-investment portfolio, so the domestic exposure $\psi_{0,t}$ is inferred residually and the equality block $\tilde{\boldsymbol{D}}$ enforces only the weight decomposition. Second, no leverage constraint is imposed on the currency exposures. The biggest difference relative to Section A.4.2 is that the fixed asset weights enter the mean-variance currency step through the linear cross-covariance term in $\tilde{\boldsymbol{c}}$, rather than through a shifted inequality constraint.

As noted by [25], the cross-covariance could alternatively be shrunk by augmenting the currency overlay problem with the fixed asset portfolio as an additional dimension. We use the direct formulation over $\boldsymbol{\psi}_t$, which keeps the currency step as both the asset program (B.4) and the Expected Shortfall currency overlay (A.11).

B.4 Joint Optimization

In the joint portfolio optimization approach, asset and currency weights are optimized simultaneously rather than sequentially. We stack them into the combined decision vector $\boldsymbol{\theta}_t = [\boldsymbol{x}_t^\top, \boldsymbol{\psi}_t^\top]^\top \in \mathbb{R}^{N+M}$ and the combined return vector $\boldsymbol{R}_{t+1} = [\boldsymbol{R}_{t+1}^{fh\top}, (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)^\top]^\top \in \mathbb{R}^{N+M}$, with joint mean $\boldsymbol{\mu} = \mathbb{E}_t[\boldsymbol{R}_{t+1}] \in \mathbb{R}^{N+M}$ and joint covariance $\boldsymbol{\Sigma} = \mathrm{Var}_t(\boldsymbol{R}_{t+1}) \in \mathbb{R}^{(N+M)\times(N+M)}$. The joint optimization problem is

$$\begin{aligned} \boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} \Big\{ & -\boldsymbol{\theta}_t^\top \boldsymbol{\mu} + \tfrac{\gamma}{2}\, \boldsymbol{\theta}_t^\top \boldsymbol{\Sigma}\, \boldsymbol{\theta}_t + \|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 \Big\}, \\ & \text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}\boldsymbol{\theta}_t\|_1 \leq \ell. \end{aligned} \tag{B.9}$$

The following matrices and vector adapt the problem to the joint optimization setting

$$\boldsymbol{q}_{N,M} = \begin{bmatrix} \mathbf{1}_N \\ \mathbf{0}_M \end{bmatrix}, \quad \boldsymbol{L} = \begin{bmatrix} \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \mathbf{0}_{M\times M} \end{bmatrix}, \quad \boldsymbol{S}_1 = \begin{bmatrix} \lambda_1^a \boldsymbol{I}_N & 0 \\ 0 & \lambda_1^c \boldsymbol{I}_M \end{bmatrix}, \quad \boldsymbol{S}_2 = \begin{bmatrix} \sqrt{\lambda_2^a}\, \boldsymbol{I}_N & 0 \\ 0 & \sqrt{\lambda_2^c}\, \boldsymbol{I}_M \end{bmatrix},$$

like in the Expected Shortfall problem (A.12). The vector $\boldsymbol{q}_{N,M}$ applies the budget constraint to the asset weights only, so that $\boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = \boldsymbol{x}_t^\top \mathbf{1}_N = 1$, leaving the currency weights as a

zero-investment overlay. The matrix $\boldsymbol{L}$ applies the leverage constraint to the asset weights only, giving $\|\boldsymbol{L}\boldsymbol{\theta}_t\|_1 = \|\boldsymbol{x}_t\|_1$. The diagonal regularization matrices $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ allow the asset and currency penalties to differ within a single objective,

$$\|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 = \lambda_1^a\|\boldsymbol{x}_t\|_1 + \lambda_1^c\|\boldsymbol{\psi}_t\|_1, \qquad \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 = \lambda_2^a\|\boldsymbol{x}_t\|_2^2 + \lambda_2^c\|\boldsymbol{\psi}_t\|_2^2.$$

The joint covariance is

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}^a & \boldsymbol{\Sigma}^{ac} \\ (\boldsymbol{\Sigma}^{ac})^\top & \boldsymbol{\Sigma}^c \end{bmatrix},$$

with the asset, currency, and cross-covariance matrices $\boldsymbol{\Sigma}^a$, $\boldsymbol{\Sigma}^c$, and $\boldsymbol{\Sigma}^{ac}$ defined as in Section B.3. The portfolio variance then expands as

$$\boldsymbol{\theta}_t^\top \boldsymbol{\Sigma}\, \boldsymbol{\theta}_t = \boldsymbol{x}_t^\top \boldsymbol{\Sigma}^a \boldsymbol{x}_t + 2\, \boldsymbol{x}_t^\top \boldsymbol{\Sigma}^{ac} \boldsymbol{\psi}_t + \boldsymbol{\psi}_t^\top \boldsymbol{\Sigma}^c \boldsymbol{\psi}_t. \tag{B.10}$$

This is the same decomposition that appears in the currency overlay step (B.7). There, however, the asset weights are fixed at $\boldsymbol{x}_t^*$, so the first term is constant and drops from the objective, while the cross term simplifies to the linear form $\gamma\, \boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}^{ac} \boldsymbol{\psi}_t$. In the joint problem both $\boldsymbol{x}_t$ and $\boldsymbol{\psi}_t$ are free. This is how thejoint approach captures the asset–currency dependence.

The joint mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ are the Bayes–Stein and Ledoit–Wolf estimators of Section B.2.2, here applied to the combined return vector $\boldsymbol{R}_{t+1}$. Because the Ledoit–Wolf shrinkage acts on the full combined covariance, the cross-covariance matrix $\boldsymbol{\Sigma}^{ac}$ is shrunk together with the asset and currency matrices. This is in contrast to the separate approach, where the cross-covariance enters the currency overlay only through the fixed asset weights $\boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}^{ac}$ and is not shrunk [25].

Applying the L_1 decomposition to all $N + M$ components of $\boldsymbol{\theta}_t$, and collecting the asset and currency parts separately, we form the augmented vector

$$\tilde{\boldsymbol{\theta}}_t = \Big[\, \boldsymbol{x}_t^\top, \quad \boldsymbol{\psi}_t^\top, \quad (\boldsymbol{\Delta}_{x,t}^-)^\top, \quad (\boldsymbol{\Delta}_{\psi,t}^-)^\top, \quad (\boldsymbol{\Delta}_{x,t}^+)^\top, \quad (\boldsymbol{\Delta}_{\psi,t}^+)^\top \,\Big]^\top \in \mathbb{R}^{3N+3M}.$$

The cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \Big[\, -\boldsymbol{\mu}^\top, \quad \lambda_1^a \mathbf{1}_N^\top, \quad \lambda_1^c \mathbf{1}_M^\top, \quad \lambda_1^a \mathbf{1}_N^\top, \quad \lambda_1^c \mathbf{1}_M^\top \,\Big]^\top,$$

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \gamma\boldsymbol{\Sigma}^a + 2\lambda_2^a \boldsymbol{I}_N & \gamma\boldsymbol{\Sigma}^{ac} & \mathbf{0}_{N\times(2N+2M)} \\ \gamma(\boldsymbol{\Sigma}^{ac})^\top & \gamma\boldsymbol{\Sigma}^c + 2\lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M\times(2N+2M)} \\ \mathbf{0}_{(2N+2M)\times N} & \mathbf{0}_{(2N+2M)\times M} & \mathbf{0}_{(2N+2M)\times(2N+2M)} \end{bmatrix}.$$

The active block of $\tilde{\boldsymbol{Q}}$ combines the joint covariance with the L_2 penalty as $\gamma\boldsymbol{\Sigma} + 2\boldsymbol{S}_2^2$, where the off-diagonal $\gamma\boldsymbol{\Sigma}^{ac}$ couples the asset and currency weights. The joint portfolio optimization problem is then the quadratic program

$$\begin{aligned} \tilde{\boldsymbol{\theta}}_t^* = \arg\min_{\tilde{\boldsymbol{\theta}}_t} \quad & \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\theta}}_t + \tfrac{1}{2}\, \tilde{\boldsymbol{\theta}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\theta}}_t, \\ \text{s.t.} \quad & \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\theta}}_t = \begin{bmatrix} \boldsymbol{q}_{N,M}^\top & \mathbf{0}_{N+M}^\top & \mathbf{0}_{N+M}^\top \\ \boldsymbol{I}_{N+M} & \boldsymbol{I}_{N+M} & -\boldsymbol{I}_{N+M} \end{bmatrix} \tilde{\boldsymbol{\theta}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_{N+M} \end{bmatrix}, \\ & \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\theta}}_t = \begin{bmatrix} \mathbf{0}_{N+M}^\top & \boldsymbol{q}_{N,M}^\top & \boldsymbol{q}_{N,M}^\top \end{bmatrix} \tilde{\boldsymbol{\theta}}_t \le b = \ell, \\ & \Delta_{x,i,t}^{\pm} \ge 0, \quad \Delta_{\psi,c,t}^{\pm} \ge 0, \quad \text{for } i = 1,\dots,N,\ c = 1,\dots,M. \end{aligned} \tag{B.11}$$

In $\tilde{\boldsymbol{D}}$, the first row enforces the asset budget constraint $\boldsymbol{x}_t^\top \mathbf{1}_N = 1$, and the remaining $N + M$ rows enforce the weight decomposition (A.6) for all components of $\boldsymbol{\theta}_t$. The single row of $\tilde{\boldsymbol{A}}$ imposes the asset-only leverage constraint (A.7), giving $\mathbf{1}_N^\top(\boldsymbol{\Delta}_{x,t}^+ + \boldsymbol{\Delta}_{x,t}^-) = \|\boldsymbol{x}_t\|_1 \leq \ell$.

The joint formulation solves a single quadratic program over $\tilde{\boldsymbol{\theta}}_t$, capturing the interaction between asset and currency exposures through the cross-covariance matrix $\boldsymbol{\Sigma}^{ac}$ in $\tilde{\boldsymbol{Q}}$. Compared with the joint Expected Shortfall program (A.13), the mean-variance program is again smaller its decision vector has dimension $3N+3M$ rather than $3N+3M+q+1$, and its only inequality is the leverage constraint. As in the Expected Shortfall case, the joint program is of higher dimension than the two separate programs of Section B.3.

All quadratic programs derived in this appendix share the same convex structure and are solvable with the same quadratic programming solver. Each extension of the program, from the base regularized single-currency formulation (B.4) to the joint multi-currency program (B.11), modifies only the covariance and cost matrices and the constraint blocks that encode the budget, leverage, and weight decomposition, without changing the program structure.

Appendix C

Transaction Costs

In this appendix, we detail the modeling of transaction costs in the context of international portfolio optimization. Following [25, 24, 13], and keeping the notation introduced in Chapter 3, we model transaction costs as proportional to the volume of traded assets and currencies, ignoring any fixed trading costs. The return of the hedged portfolio is expressed as the sum of unhedged asset returns and gains from forward currency positions

$$R^h_{P,t+1} = \sum_{i=1}^{N} x_{i,t} R^u_{i,t+1} + \sum_{c=1}^{M} \phi_{c,t}(f_{c,t} - e_{c,t+1}). \tag{3.2}$$

Let τ^{asset}_{t+1} and τ^{curr}_{t+1} denote the total transaction costs associated with asset and currency trades, respectively. The portfolio return net of transaction costs is then given by

$$R^{h,net}_{P,t+1} = R^h_{P,t+1} - \tau^{asset}_{t+1} - \tau^{curr}_{t+1}. \tag{C.1}$$

The total transaction costs are computed as the sum of the costs incurred from trading individual assets and currencies, weighted by the trading volumes

$$\tau^{asset}_{t+1} = \sum_{i=1}^{N} \tau^{asset}_{i,t+1} \left(x_{i,t+1} - x^-_{i,t+1} \right),$$

$$\tau^{curr}_{t+1} = \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \phi_{c,t+1} + \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \left(w_{c,t+1} - w^-_{c,t+1} \right),$$

where $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$ are the per-unit transaction costs for asset i and currency c. The terms $x^-_{i,t+1}$ and $w^-_{c,t+1}$ denote the asset weights and implied currency weights immediately prior to rebalancing at time $t+1$ (i.e., the drifted weights).

Asset transaction costs are incurred only when rebalancing individual asset holdings. Currency transaction costs, however, arise from two sources, adjustments in forward hedge positions (assuming the forward contract maturity matches the rebalancing frequency, the old contract expires and a new contract of full notional size $\phi_{c,t+1}$ must be established at each rebalancing date) and currency exchanges required to fund purchases of assets denominated in different currencies ($w_{c,t+1} - w^-_{c,t+1}$). If the portfolio is rebalanced more frequently than the forward contract maturity, existing forward contracts are kept open and only incremental adjustments are traded. In this case, the forward transaction cost term is modeled as the absolute difference, replacing $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} \phi_{c,t+1}$ with $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} (\phi_{c,t+1} - \phi^-_{c,t+1})$, where $\phi^-_{c,t+1}$ represents the drifted hedge weight. Following [25], these costs are aggregated at the currency level, operating under

the assumption that proceeds from selling assets denominated in currency c can fund the purchase of other assets in the same currency without incurring currency conversion fees.

To define the individual per-unit costs $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$, we consider an asset i denominated in currency c_i. Let $P^{ask}_{i,t}$, $P^{bid}_{i,t}$, and $P_{i,t}$ represent the ask, bid, and mid prices of the asset, and let $S^{ask}_{c_i,t}$, $S^{bid}_{c_i,t}$, and $S_{c_i,t}$ represent the corresponding currency prices. If asset i is purchased at time t and sold at time $t+1$, its realized log return is given by

$$\begin{aligned}
\log\left(\frac{P^{bid}_{i,t+1}S^{bid}_{c_i,t+1}}{P^{ask}_{i,t}S^{ask}_{c_i,t}}\right) &= \log\left(\frac{\left(P_{i,t+1}-\frac{\delta^{asset}_{i,t+1}}{2}\right)\left(S_{c_i,t+1}-\frac{\delta^{curr}_{c_i,t+1}}{2}\right)}{\left(P_{i,t}+\frac{\delta^{asset}_{i,t}}{2}\right)\left(S_{c_i,t}+\frac{\delta^{curr}_{c_i,t}}{2}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}\left(1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}\right)S_{c_i,t+1}\left(1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}\right)}{P_{i,t}\left(1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}\right)S_{c_i,t}\left(1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)+\log\left(\frac{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}\right)+\log\left(\frac{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)-\log\left(\frac{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}\right)-\log\left(\frac{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}\right),
\end{aligned}$$

where δ^{asset}_i and $\delta^{curr}_{c_i}$ are the bid-ask spreads. Comparing this to equation (C.1), the transaction cost is precisely the difference between the actual realized log return and mid-price returs for the asset and currency.

Defining half of the bid-ask spreads as

$$\chi^{asset}_{i,t} = \frac{\overbrace{P^{ask}_{i,t}-P^{bid}_{i,t}}^{\delta^{asset}_i}}{2P_{i,t}} \quad \text{and} \quad \chi^{curr}_{c,t} = \frac{\overbrace{S^{ask}_{c,t}-S^{bid}_{c,t}}^{\delta^{curr}_c}}{2S_{c,t}},$$

the per-unit transaction costs for trading asset i and currency c become

$$\tau^{asset}_{i,t+1} = \begin{cases} \log\left(\frac{1+\chi^{asset}_{i,t+1}}{1-\chi^{asset}_{i,t}}\right) & \text{if } x_{i,t+1} > x^{-}_{i,t+1}, \\ \log\left(\frac{1-\chi^{asset}_{i,t}}{1+\chi^{asset}_{i,t+1}}\right) & \text{otherwise,} \end{cases}$$

and

$$\tau^{curr}_{c,t+1} = \begin{cases} \log\left(\frac{1+\chi^{curr}_{c,t+1}}{1-\chi^{curr}_{c,t}}\right) & \text{if } \phi_{c,t+1} > 0 \text{ or } w_{c,t+1} > w^{-}_{c,t+1}, \\ \log\left(\frac{1-\chi^{curr}_{c,t}}{1+\chi^{curr}_{c,t+1}}\right) & \text{otherwise.} \end{cases}$$

For our empirical analysis, we adopt the simplifying assumption that proportional bid-ask spreads remain constant both over time and across the cross-sections of both assets and currencies. We set the half-spreads to $\chi^{asset} = 0.001$ (10 bps) for all equities and $\chi^{curr} = 0.0001$ (1 bp) for all currencies, consistent with empirical finidngs in [8, 19, 17, 6]. Under these assumptions, the transaction costs simplify to constant values of $\tau^{asset} \approx 0.002$ and $\tau^{curr} \approx 0.0002$.

Appendix D

Hyperparameter Sensitivity and Portfolio Sparsity Analysis

In this appendix, we justify the bounded hyperparameter grids over which the regularization penalties are selected in the empirical analysis of Chapter 5. In Section D.1 we describe the sensitivity analysis used to determine these bounds, following a similar approach as [24].

TODO: Summarize the sparsity analysis and its purpose.

D.1 Hyperparameter Sensitivity Analysis

The regularized portfolio optimization problems include four regularization parameters, the asset penalties λ_1^a, λ_2^a and the currency penalties λ_1^c, λ_2^c. These are selected by grid search on a validation sample, which requires setting a bounded interval for each. The bounds are not arbitrary. As a penalty grows, the optimal weights approach a limiting allocation, and beyond some threshold a further increase leaves the solution essentially unchanged. A grid whose upper bound lies past that threshold is sufficient, in the sense that extending it further would only increase the computational cost of the grid search.

To locate this threshold we follow [24] and define, for each penalty $\lambda \in \{\lambda_1^a, \lambda_2^a, \lambda_1^c, \lambda_2^c\}$, an increasing sequence $(\lambda_n)_{n\in\mathbb{N}}$ with $\lambda_0 = 0$. Let $\boldsymbol{x}_t(\lambda_n)$ and $\boldsymbol{\psi}_t(\lambda_n)$ denote the optimal asset and currency overlay weights when only λ is active, that is, when the remaining three penalties are held at zero. The marginal effect of an incremental increase in regularization is measured in the supremum norm

$$d_n^a = \|\boldsymbol{x}_t(\lambda_n) - \boldsymbol{x}_t(\lambda_{n-1})\|_\infty \quad \text{for } \lambda \in \{\lambda_1^a, \lambda_2^a\},$$
$$d_n^c = \|\boldsymbol{\psi}_t(\lambda_n) - \boldsymbol{\psi}_t(\lambda_{n-1})\|_\infty \quad \text{for } \lambda \in \{\lambda_1^c, \lambda_2^c\}.$$

Tthe two penalty types approach their limits in different ways. Under an L_1 penalty the weights move to the corner of the feasible set, at which point the penalty term becomes constant in the weights and doesnt influence the solution. An L_2 penalty instead shrinks the weights smoothly toward the origin and never reaches a corner, so its marginal effect decays gradually without ever vanishing. Such behavior is well documented in [**?**]. For this reason we report alongside d_n the marginal effect on portfolio exposure, $\|\boldsymbol{x}_t\|_1$ for the asset penalties and $\|\boldsymbol{\psi}_t\|_1$ for the currency penalties. Exposure is the economically interpretable quantity, and is the one that we use to determine the upper bound of the grid for the L_2 penalties.

The analysis is done on the first estimation window spanning July 1990 to June 2000. To account for sampling error, each reported quantity is averaged over 5 bootstraped replicates of the estimation window. The results are reported for the small investable universe.

D.1.1 Expected Shortfall Analysis

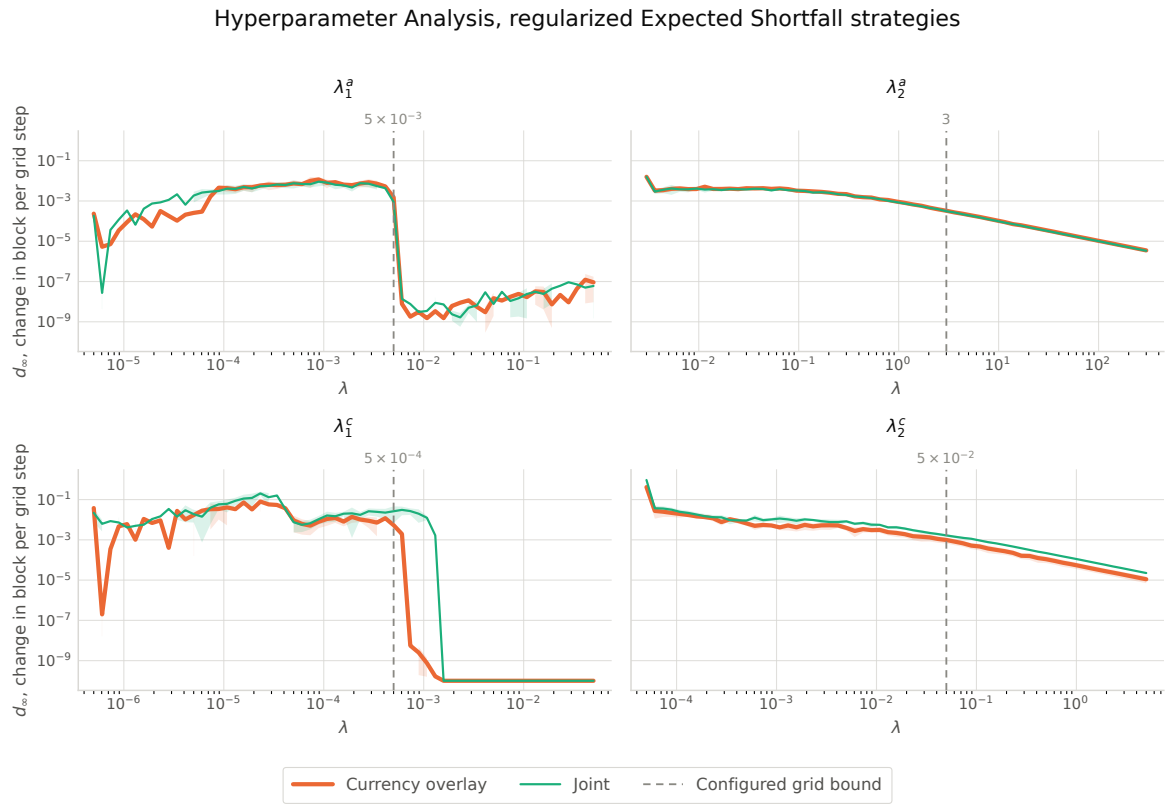


Figure D.1: Marginal effect of each regularization penalty, for the regularized Expected Shortfall currency overlay and joint optimizers. The dashed line marks the upper bound of the hyperparameter grid reported in Table D.1. The upper plots report d_n^a and the lower plots d_n^c. Estimates are averaged over five bootstrap replicates of the first estimation window on the small investment universe.

Figure D.1 and Figure D.2 report the marginal effect and the portfolio exposure. The difference between the two penalty types is clearest in the marginal effect plots. The L_1 penalties collapse abruptly near their bound, whereas the L_2 penalties decay smoothly throughout, which is why their bounds are determined by the exposure analysis instead.

For both asset penalties, exposure falls from the leverage limit in the unpenalized problem to 1. The budget constraint $\boldsymbol{x}_t^\top \mathbf{1}_N = 1$ implies $\|\boldsymbol{x}_t\|_1 \geq 1$, with the equality reached when the portfolio holds no short positions/ is long only. Once it is reached, the penalty term is constant in $\boldsymbol{x}_t$ and can no longer influence the solution.

The overlay weights carry neither a budget nor a leverage constraint, so $\boldsymbol{\psi}_t$ shrinks to zero.

On the basis of this analysis we fix the ranges reported in Table D.1. Each range is discretised on a log-spaced grid that includes the zero endpoint, which corresponds to the unregularized optimization problem.

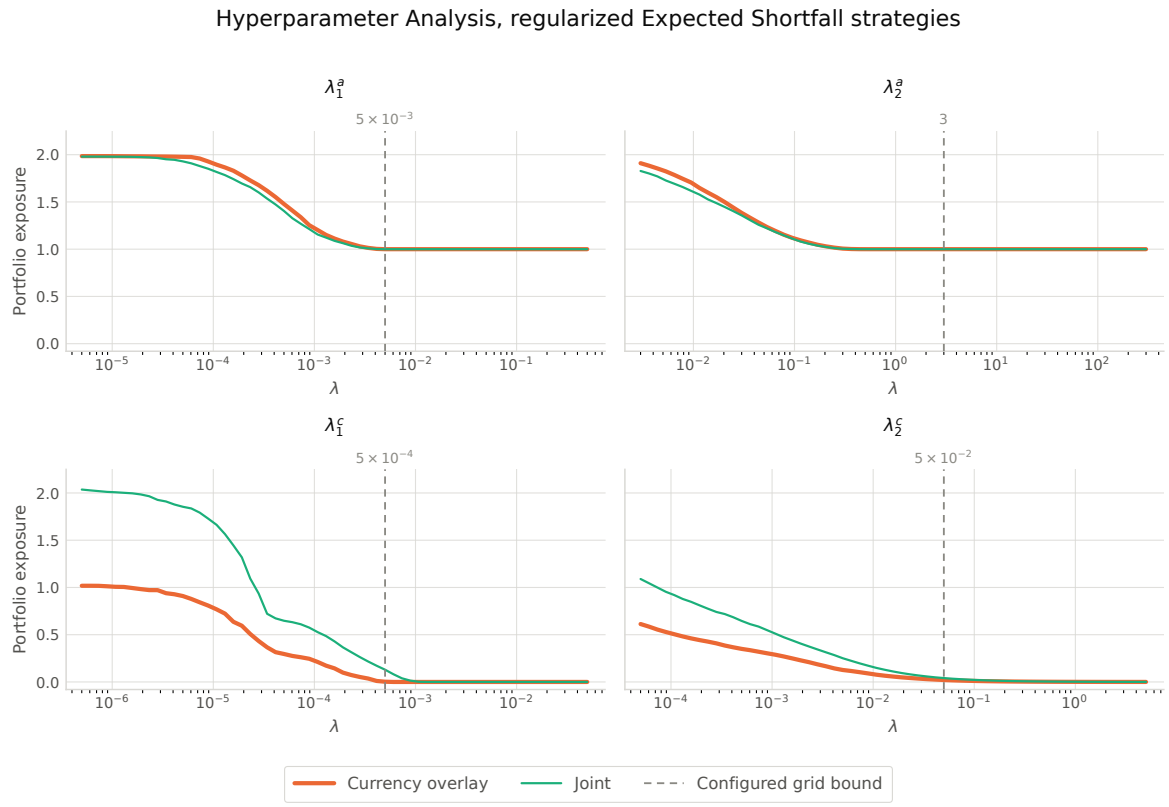


Figure D.2: Portfolio exposure as a function of each regularization penalty, for the regularized Expected Shortfall currency overlay and joint optimizers. The dashed line marks the upper bound of the validation grid reported in Table D.1. The upper panels report $\|\boldsymbol{x}_t\|_1$ and the lower panels $\|\boldsymbol{\psi}_t\|_1$. Estimates are averaged over five bootstrap replicates of the first estimation window on the small investment universe.

Hyperparameter	Description	Range
λ_1^a	Asset L_1 penalty	$[0,\ 5 \times 10^{-3}]$
λ_2^a	Asset L_2 penalty	$[0,\ 3]$
λ_1^c	Currency L_1 penalty	$[0,\ 5 \times 10^{-4}]$
λ_2^c	Currency L_2 penalty	$[0,\ 5 \times 10^{-2}]$

Table D.1: Hyperparameter ranges for the regularized Expected Shortfall optimizers. Each penalty is searched over a log-spaced grid spanning the reported range, where the lower endpoint corresponds to the unregularized problem and the upper endpoint is the bound established in Section D.1.1.

D.1.2 Mean-Variance Analysis

D.1.3 Mean Absolute Deviation Analysis

D.2 Portfolio Sparsity Analysis

D.2.1 Sparsity of the Regularized Expected Shortfall Optimizers

Figure D.3 reports the share of the investable universe held at each rebalance date.

Regularization stabilizes the weight vector conditional on the penalty coefficients, but those coefficients are themselves re-selected at every rebalance date. The number of active positions is sensitive to this recalibration, an L1-dominant selection sets weights to zero and yields a concentrated portfolio, whereas an L2-dominant selection shrinks all weights toward the equally

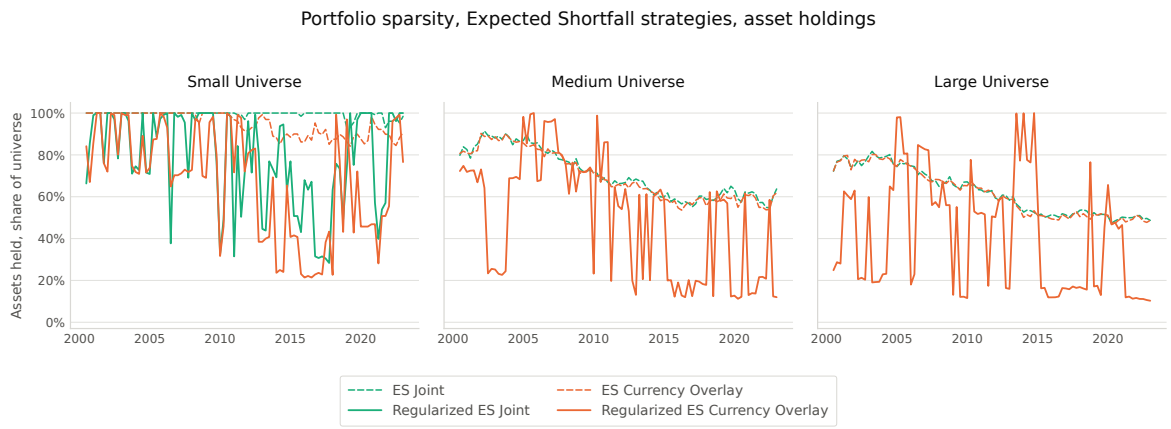


Figure D.3: Sparsity of the regularized Expected Shortfall currency overlay and joint optimizers.

weighted portfolio. The oscillation therefore reflects the validation criterion alternating between these two regimes across quarters, not instability in the optimizer itself. The unregularized problem has no such degree of freedom. Its support is determined primarily by the leverage limit.

D.2.2 Sparsity of the Regularized Mean-Variance Optimizers

D.2.3 Sparsity of the Regularized Mean Absolute Deviation Optimizers

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